



PFA INTERNSHIP REPORT

2nd Year Computer Engineering

Automating the IT Service Request Lifecycle

(Request Fulfillment) Using ServiceNow

Internship Period

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I would like to express my sincere gratitude to the management of DXC Technology Morocco for giving me the opportunity to complete my PFA internship remotely and to contribute to a professional project within a real working environment. This experience allowed me to put the knowledge acquired during my studies into practice while developing new technical and professional skills.

I would especially like to thank Houda AJDIR, my company supervisor, for her guidance, availability, and continuous support throughout the internship. Despite the remote nature of the internship, her explanations, feedback, and technical guidance were essential to understanding the project requirements, addressing the challenges encountered, and progressing effectively throughout the different stages of the work.

The remote nature of the internship also encouraged me to work independently, organize my tasks effectively, manage my time, and take responsibility for the progress of my work. I am grateful to my supervisor for the trust and autonomy given to me throughout this experience.

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This internship has been a valuable step in my academic and professional development, and I am grateful for the opportunity and experience it provided me.

Abstract

This report presents the work carried out during my PFA internship at **DXC Technology Morocco**. The internship was conducted remotely and was divided into two main phases. The first month was dedicated to professional training through a ServiceNow System Administrator course, which provided me with the fundamental knowledge required to understand and work with the ServiceNow platform. The second month focused on applying this knowledge through the design and implementation of a Service Request Fulfillment solution.

The project aimed to structure and automate the management of IT service requests in order to improve their processing, traceability, assignment, and overall fulfillment. Several service catalog items were designed and configured, including User Account Creation, Enterprise VPN Access, Software Installation, IT Hardware Requests, and Password Reset. The implementation involved defining request forms, approval processes, fulfillment tasks, assignment rules, notifications, and automated closure. Service Level Agreements (SLAs) were also configured to support service monitoring and ensure compliance with defined service targets.

The project was developed within the ServiceNow environment through a structured process involving requirements analysis, configuration, testing, validation, and deployment preparation. This internship provided me with practical experience in ServiceNow administration and configuration, IT Service Management, workflow automation, and requirements analysis. It also strengthened my ability to work independently, manage my time effectively, and apply newly acquired technical knowledge to a practical professional project.

Keywords: ServiceNow, IT Service Management (ITSM), Service Request Fulfillment, Service Catalog, Workflow Automation, Service Level Agreement (SLA), Business Rules

List of abbreviations

abbreviation	Meaning
IT	Information Technology
ITSM	Information Technology Service Management
ITIL	Information Technology Infrastructure Library
UI	User Interface
SLA	Service Level Agreement
aPaaS	Application Platform as a Service
SaaS	Software as a Service
UAT	User Acceptance Testing
CMDB	Configuration Management Database
PDI	Personal Developer Instance

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Chapter 1

General Context of the Internship

1.1 Introduction

this chapter presents the general context of my PFA Internship at **DXC Technology Morocco**. It begins with an overview of the host organization and its activities, followed by a presentation of the internship context and working environment. The chapter then introduces the main project carried out during the internship, including its background, problem statement, objectives, and scope. Finally, the methodology and general approach adopted throughout the project are presented.

1.2 Presentation of the Host Organization

1.2.1 DXC Technology Worldwide

DXC Technology is a global technology services company that helps organizations manage, modernize, and transform their information technology environments. The company provides a broad range of IT services and solutions covering applications, cloud and infrastructure, cybersecurity, analytics, workplace services, and business process services.

DXC Technology serves organizations across multiple industries and geographical regions, supporting them in addressing complex technological and operational challenges. Its activities combine consulting, technology implementation, systems integration, and managed services to support the transformation and continuous operation of enterprise IT environments.



Figure 1.1: DXC Technology Logo

1.2.1.1 Computer Sciences Corporation (CSC)

Computer Sciences Corporation (CSC) was an American information technology company specializing in technology consulting, systems integration, outsourcing, and IT services. Founded in 1959, CSC developed extensive expertise in helping organizations design, implement, and manage complex information systems.

Over the years, CSC established an international presence and provided technology services to organizations across various industries. Its experience in consulting, application services, and enterprise IT management later became an important part of the capabilities brought together within DXC Technology.



Figure 1.2: CSC logo

1.2.1.2 Hewlett Packard Enterprise (HPE)

Hewlett Packard Enterprise (HPE) is an American technology company focused on enterprise computing, infrastructure, networking, storage, and related IT services. HPE was established in 2015 following the separation of Hewlett-Packard Company into two independent organizations.

The Enterprise Services business of HPE provided organizations with services related to infrastructure management, outsourcing, applications, and enterprise technology environments. This business subsequently became one of the two major components involved

in the creation of DXC Technology.



Figure 1.3: HPE logo

1.2.1.3 Creation of DXC Technology

DXC Technology was created in 2017 through the merger of Computer Sciences Corporation (CSC) and the Enterprise Services business of Hewlett Packard Enterprise (HPE). The merger combined the technological expertise, service capabilities, and international presence of both organizations to establish a new global IT services company.

DXC Technology officially began operations on April 1, 2017. Since its creation, the company has continued to develop its expertise in areas such as applications, cloud and infrastructure, cybersecurity, analytics, workplace services, and business process services.

ABOUT DXC – BORN IN 2017 BUT NOT NEW

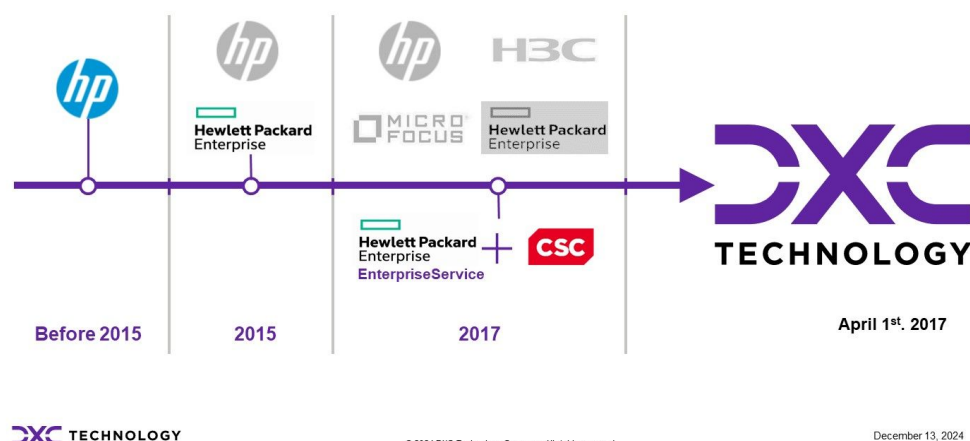


Figure 1.4: Evolution leading to the creation of DXC Technology

1.2.2 DXC Technology Morocco

DXC Technology Morocco is the Moroccan entity of DXC Technology and operates within the company's global network of technology services. The organization supports major public and private-sector organizations in their digital transformation initiatives and provides a range of technology and business services.

1.2.2.1 Overview of DXC Technology Morocco

Item	Information
Company	DXC Technology Morocco
Location	Technopolis, Rabat, Morocco
Established	2007
Current name	DXC Technology Morocco since 2017
Parent organization	DXC Technology
Local partner	Caisse de Dépôt et de Gestion (CDG)
Main activity	Information Technology Services

Table 1.1: DXC Technology Morocco at a Glance

1.2.2.2 History of DXC Technology Morocco

The history of DXC Technology Morocco dates back to 2007, when the Moroccan entity was established through a partnership between Electronic Data Systems (EDS) and the Caisse de Dépôt et de Gestion (CDG). The organization initially operated from a 450 m² site with 36 employees and began providing IT services to support CDG's activities.

In August 2008, EDS was acquired by Hewlett-Packard (HP), marking an important step in the development of the Moroccan entity. In 2009, the organization integrated HP's activities and changed its branding to HP CDG IT Services Maroc. During the same period, it obtained ISO 9001 certification.

The organization continued to expand during the following years. In 2011, it was recognized as the fifth-best employer. In 2012, the organization moved to a new site designed to accommodate up to 1,000 employees and was recognized as the third-best employer. By 2014, it had reached the second position in the best-employer ranking.

These milestones illustrate the progressive development of the Moroccan organization in terms of infrastructure, workforce, and recognition as an employer.

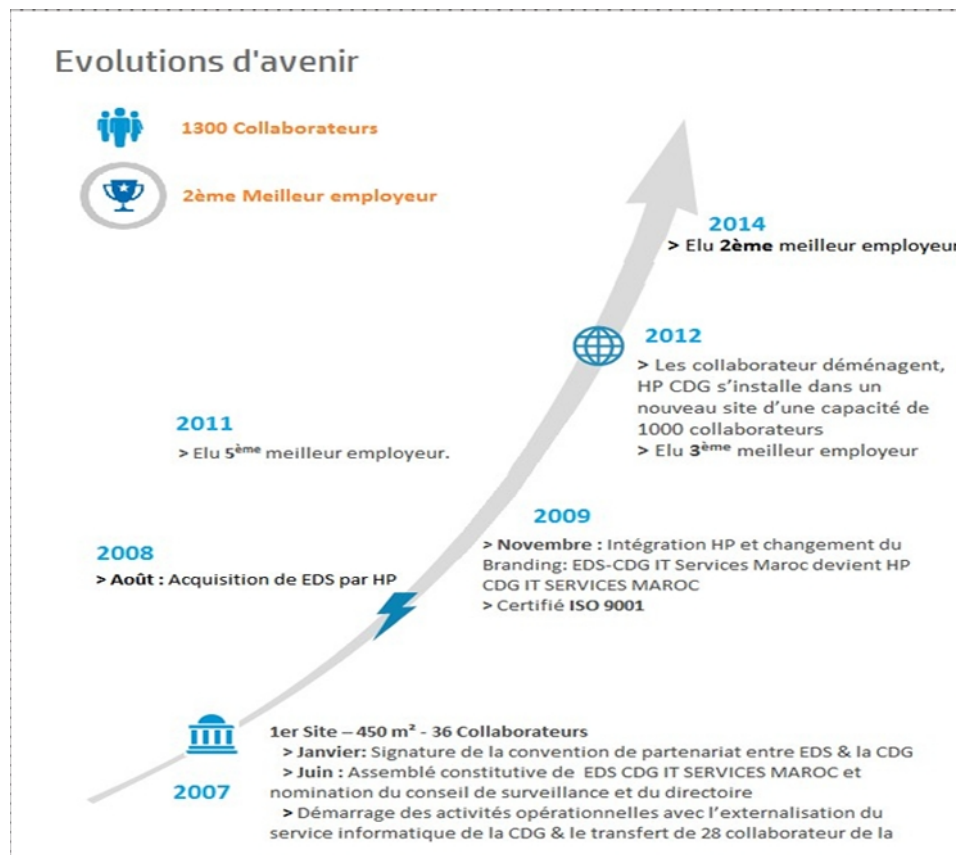


Figure 1.5: Historical development of DXC Technology Morocco

Following these developments, the organization underwent another major transformation in 2017 following the global creation of DXC Technology through the combination of Computer Sciences Corporation (CSC) and Hewlett Packard Enterprise's Enterprise Services business. The Moroccan entity subsequently adopted the DXC Technology name, becoming part of DXC Technology's global organization.

1.2.2.3 DXC Technology Success Story

The following figure presents the success story of DXC Technology and the evolution of the number of employees since 2007.



Figure 1.6: DXC Technology Success Story

1.2.2.4 Business Areas and Services

DXC Technology Morocco provides a diversified range of technical and functional services organized around several major areas of expertise. These areas combine technology capabilities with business-oriented services to address the requirements of organizations across different sectors.

- **Data Center:** Hosting, technical support, infrastructure supervision, and deployment of cloud solutions for efficient data management.
- **Cloud Infrastructure:** Design and modernization of hybrid cloud environments and migration of traditional platforms toward evolving cloud architectures.
- **Applications:** Integration, migration, modernization, and maintenance of enterprise applications.
- **ServiceNow:** Integration, customization, and optimization of business and IT processes through the ServiceNow platform, particularly in areas such as IT Service Management (ITSM) and IT Operations Management (ITOM).
- **Workplace and Mobility:** Secure access to enterprise resources through endpoint management, device management, and workplace virtualization solutions.
- **Business Process Services:** Management and support of business processes, including customer service and automated request processing.

- **Cybersecurity:** Security assessment, risk management, monitoring, incident management, and deployment of security solutions.
- **Business Intelligence and Analytics:** Development of analytical platforms and data solutions to support reporting, analysis, and decision-making.

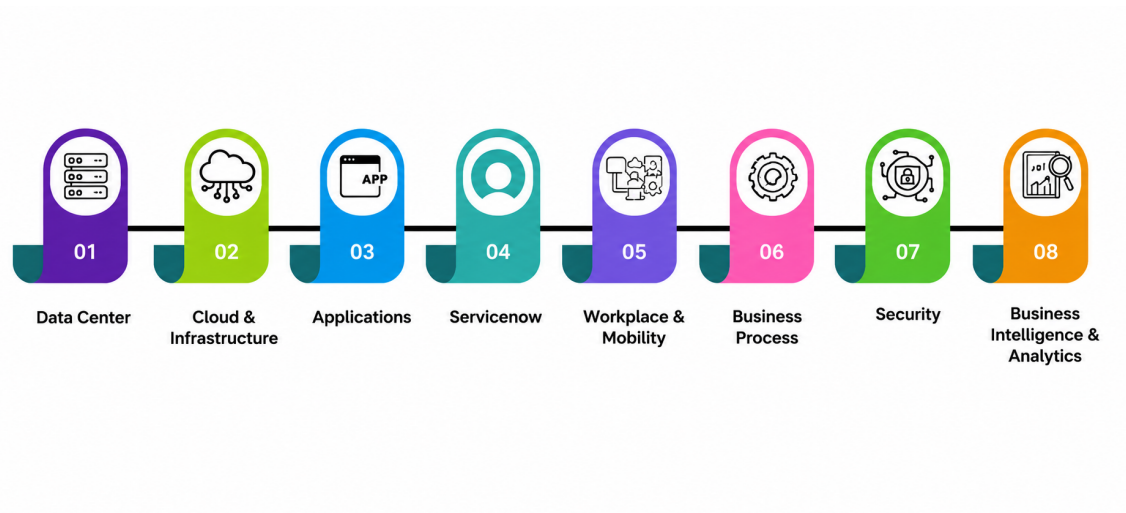


Figure 1.7: Main business areas and services of DXC Technology Morocco

1.2.2.5 Organizational Structure

DXC Technology Morocco is organized into different functional and operational structures that support the delivery of technology services to its customers. These structures include management and support functions as well as specialized service lines responsible for delivering technical and business solutions.

Figure 1.8 presents an overview of the organizational structure of DXC Technology Morocco.

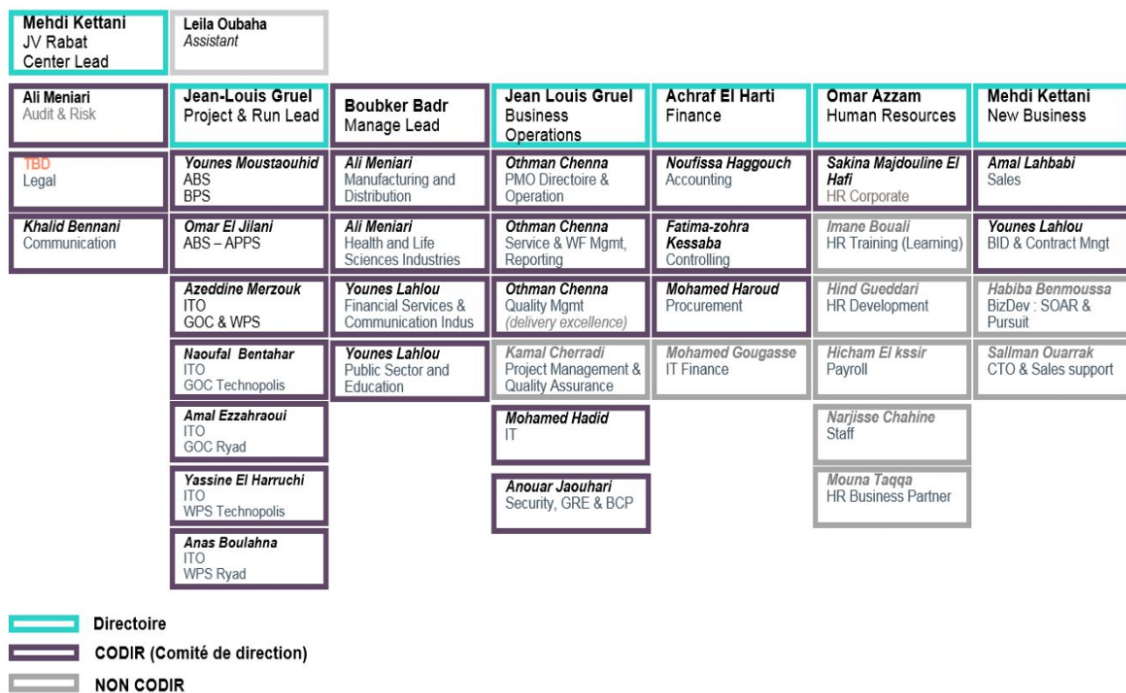


Figure 1.8: Organizational structure of DXC Technology Morocco

1.2.2.6 Clients and Strategic Partners

DXC Technology Morocco serves organizations operating across a wide range of industries, including manufacturing, retail, banking and finance, telecommunications, energy, healthcare, transportation, tourism, the public sector, and real estate.

The diversity of its customer base reflects the broad range of technology and business services provided by the organization.



Figure 1.9: Examples of DXC Technology Morocco's clients

DXC Technology Morocco also works with technology partners and maintains industry certifications that support the quality, security, and reliability of its services. These partnerships and certifications contribute to the company's ability to deliver solutions that meet technological and operational requirements.



Figure 1.10: Selected partners and certifications of DXC Technology Morocco

1.3 Internship Context

1.3.1 Internship Framework

As part of my academic curriculum, I completed a two-month PFA internship at **DXC Technology Morocco**. The internship was conducted entirely remotely and focused on the discovery and practical application of the ServiceNow platform.

The internship was supervised by **Houda AJDIR**, who provided guidance, clarifications, and feedback throughout the internship. The work was carried out independently, with regular communication with the company supervisor regarding the progress and requirements of the assigned work.

1.3.2 Internship Objectives

The main objective of this internship was to develop practical knowledge of the **ServiceNow** platform and gain experience in its administration and configuration within a professional IT environment.

More specifically, the internship aimed to:

- Acquire a solid understanding of the ServiceNow platform and its main administration concepts through the System Administrator training.
- Develop an understanding of **IT Service Management (ITSM)** concepts and their implementation within ServiceNow.
- Apply the knowledge acquired during the training to a practical ServiceNow project.

- Design and configure a structured **Service Request Fulfillment** solution for managing IT service requests.
- Gain practical experience with ServiceNow configuration components, including forms, workflows, approval processes, assignment rules, notifications, and business rules.
- Understand how automation can be used to improve the processing, traceability, and fulfillment of IT service requests.
- Develop the ability to work independently, organize tasks effectively, and manage a technical project in a remote working environment.

1.3.3 Remote Working Environment

The internship was conducted entirely remotely, without any physical presence at the company's premises. This working arrangement required me to organize my work independently and manage my time according to the objectives and requirements of the internship.

Throughout the internship, the main channel of communication was with my company supervisor, **Houda AJDIR**. She provided the necessary guidance, clarifications, and feedback regarding the training and the assigned project.

The remote nature of the internship also required a high degree of autonomy and personal organization. Tasks were carried out independently, while progress and questions were communicated to the supervisor when necessary. This environment allowed me to develop my ability to manage my workload, solve problems independently, and maintain progress throughout the different stages of the internship.

1.3.4 Internship Phases

The internship was organized into two main phases, each with a specific objective. The first phase focused on acquiring the necessary knowledge of the ServiceNow platform, while the second phase consisted of applying this knowledge to a practical project.

Phase	Duration	Main Activities
Phase 1	First Month	ServiceNow System Administrator training, covering the fundamental concepts of the platform and its administration and configuration.
Phase 2	Second Month	Design and implementation of the Service Request Fulfillment project, including the configuration and testing of the required ServiceNow components.

Table 1.2: Organization of the Internship

During the first phase, the focus was on building a solid foundation in ServiceNow administration. The training provided the knowledge required to understand the platform and its main configuration concepts.

During the second phase, the knowledge acquired during the training was applied to the development of the Service Request Fulfillment solution. This phase involved analyzing the requirements, designing the service processes, configuring the ServiceNow components, and testing the resulting solution.

1.4 Project Context

1.4.1 Background (Service Request Management)

In modern enterprise environments, Information Technology Service Management (ITSM) is critical for aligning IT services with business needs. Within the ITSM framework, Request Fulfillment is the core process responsible for managing the lifecycle of all standard service requests from users. These requests typically include access to applications, software installations, hardware provisioning, and routine information queries. An effective Request Fulfillment process ensures that employees have access to the IT resources they need to perform their daily tasks efficiently, while the organization maintains proper authorization, compliance, and security controls.

1.4.2 Problem Statement

Despite the critical nature of IT services, many organizations continue to rely on manual or fragmented processes to handle user requests. When requests are submitted via unstructured channels such as emails, phone calls, or direct messages, several operational issues arise:

- **Lack of Traceability:** It is difficult to track the status of a request, leading to poor visibility for both the end-user and the IT support team.
- **Delayed Processing:** Manual routing of requests for managerial or security approvals often creates bottlenecks, increasing the time required to fulfill the request.
- **Assignment Errors:** Without an automated routing mechanism, requests are frequently assigned to the wrong fulfillment group, causing further delays.
- **Absence of Metrics:** The inability to track request fulfillment times makes it difficult to measure performance against Service Level Agreements (SLAs) or identify areas for continuous improvement.

These challenges highlight the need for a structured and automated request fulfillment process that ensures requests are properly captured, authorized, assigned, and monitored throughout their lifecycle.

1.4.3 Project Objectives

To address these operational inefficiencies, this project aims to design and implement an automated Request Fulfillment solution using the ServiceNow platform. The primary objectives are :

- **Centralize Requests:** Provide a single, structured portal (Service Catalog) where users can easily and intuitively submit IT requests.
- **Automate Workflows:** Replace manual approval and routing steps with automated workflows to ensure requests are immediately directed to the appropriate managers and fulfillment teams.

- **Standardize Data Collection:** Use dynamic request forms to ensure all necessary information is captured at the time of submission, eliminating back-and-forth communication.
- **Enforce SLAs:** Configure Service Level Agreements to monitor fulfillment times and trigger automated escalations when targets are at risk.
- **Improve Traceability:** Provide visibility into the status and progress of requests throughout their fulfillment lifecycle.

1.4.4 Project Scope

The scope of this project encompasses the end-to-end configuration of five specific Service Catalog items that represent common and critical IT request :

1. **User Account Creation:** Automating the approval and task generation for provisioning Active Directory, email, and application accounts for new employees.
2. **Enterprise VPN Access:** Structuring the security and network approval chains required for secure remote access.
3. **Software Installation:** Managing software requests including automated license verification and installation tasks.
4. **IT Hardware Requests:** Automating the process from stock verification and procurement approval to delivery and configuration.
5. **Password Reset:** Implementing a secure, streamlined process for routine account recovery.

The technical scope involves leveraging ServiceNow's core capabilities, including the Service Catalog, Flow Designer for process automation, Business Rules and Client Scripts for backend logic, with UI Policies, User Criteria for access control, and SLA definitions.

1.4.5 Project Deliverables

The main deliverable of the project was a configured ServiceNow Request Fulfillment solution covering five IT service catalog items. The solution included structured request

forms, approval and fulfillment processes, assignment mechanisms, notifications, and Service Level Agreements.

The project also resulted in configuration artifacts that could be packaged using ServiceNow Update Sets to support the controlled migration of the developed solution between environments.

1.5 Methodological Approach

To ensure the successful delivery of the Request Fulfillment solution, a structured and iterative methodological approach was adopted. The project was executed through the following key phases:

- **Requirements Analysis:** Defining the specific inputs, variables, approval chains, and fulfillment tasks required for each of the five catalog items. This phase involved translating business requirements into technical ServiceNow specifications.
- **Platform Configuration and Development:** Utilizing a ServiceNow Development instance to build the solution. This involved creating Catalog Items, defining variables and UI Policies for dynamic forms, and utilizing Flow Designer to build the underlying automation logic.
- **Testing and Validation:** Conducting rigorous testing of each workflow to ensure approvals trigger correctly, tasks are assigned to the right fulfillment groups based on the request type, and SLAs attach properly based on ticket priority.
- **Deployment Preparation:** Packaging the configurations into ServiceNow Update Sets to ensure a safe, structured, and traceable migration of the developed features from the Development environment to Testing (UAT) and Production environments.

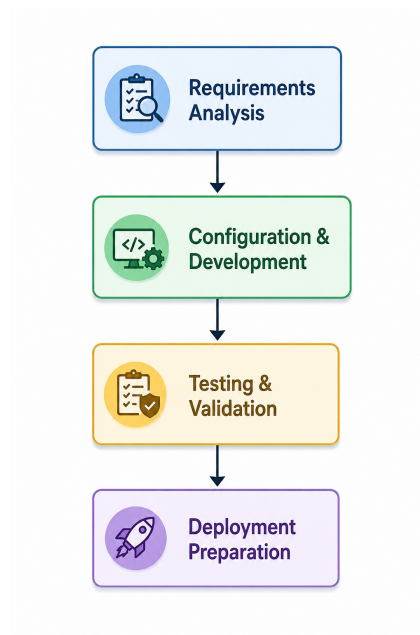


Figure 1.11: Methodological Approach Phases

1.6 Conclusion

This chapter presented the context of the internship at DXC Technology Morocco, the Service Request Fulfillment project, its objectives and scope, and the methodological approach adopted. The following chapter introduces the ServiceNow platform, IT Service Management (ITSM), and the main technical concepts used in the project.

Chapter 2

Theoretical and Technical Background

2.1 Introduction

This chapter presents the theoretical and technical concepts required to understand the Service Request Fulfillment solution developed during the internship. It introduces the ServiceNow platform, its architecture, and its core capabilities. Furthermore, it explores the principles of Information Technology Service Management (ITSM) and the role of ITIL best practices in service request management. Finally, the chapter details the main ServiceNow components used to design, automate, and prepare the deployment of the solution, including the Service Catalog, Flow Designer, Service Level Agreements (SLAs), and the ServiceNow development lifecycle.

2.2 ServiceNow Platform



Figure 2.1: ServiceNow Logo

2.2.1 Overview of ServiceNow

ServiceNow is a cloud-based enterprise platform that provides Software as a Service (SaaS) and Application Platform as a Service (aPaaS) solutions. It is designed to support the management and automation of enterprise processes and services. Initially focused on Information Technology Service Management (ITSM), the platform has progressively ex-

panded its capabilities to cover different areas of an organization, including IT operations, security, human resources, customer service, and business processes.

The platform provides a centralized environment in which organizations can define processes, manage requests, automate tasks, and monitor their execution. Its configurable architecture also allows organizations to adapt the platform to their specific business and operational requirements.

Founded	2004
Founded by	Fred Luddy
Headquarters	Santa Clara, California, USA
Logo	
Website	ServiceNow.com

Table 2.1: ServiceNow Overview

2.2.2 ServiceNow Architecture

ServiceNow is based on a cloud architecture designed to provide each customer with an isolated and configurable environment. One of the main characteristics of the platform is its **multi-instance architecture**, in which each customer is provided with a dedicated instance.

Unlike a traditional multi-tenant architecture, where multiple customers share the same application environment, the multi-instance approach provides a greater level of isolation between customer environments. Each instance contains its own application, middleware, and database resources, while the underlying physical infrastructure can be shared.

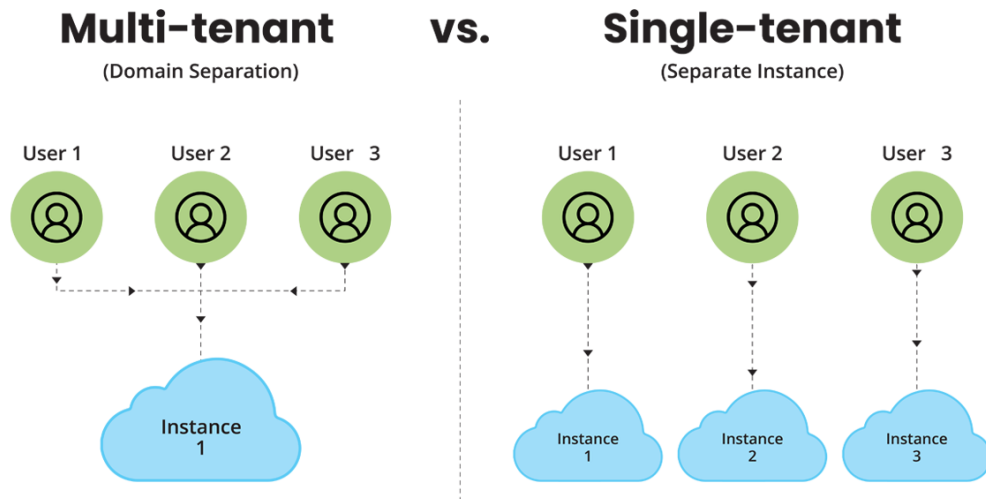


Figure 2.2: Multi Tenant vs Single Tenant

This architecture provides several advantages, particularly in terms of data isolation, configuration flexibility, and platform management. changes or configurations made within one instance are isolated from other customer instances, allowing organizations to adapt the platform to their specific requirements.

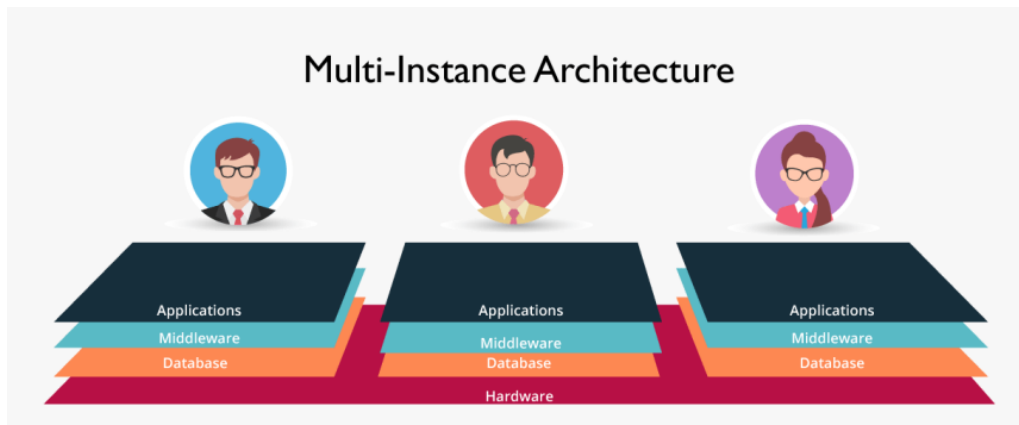


Figure 2.3: Multi-Instance Architecture of ServiceNow

2.2.3 ServiceNow Applications and Modules

ServiceNow provides a wide range of applications and modules designed to support different organizational processes and address specific enterprise needs :

- **ITSM (IT Service Management):** The core module focused on delivering and managing IT services, encompassing incident, problem, change, and request man-

agement.

- **ITOM (IT Operations Management):** Focuses on infrastructure visibility, service health, and cloud provisioning.
- **ITBM (IT Business Management):** Aligns IT investments with business goals through project portfolio management and financial tracking.
- **Security Operations:** Integrates security incident response and vulnerability management directly into IT workflow.

This project is built primarily on the ITSM module, leveraging its Service Catalog and workflow automation capabilities.

2.2.4 ServiceNow Development and Configuration

ServiceNow is a low-code/no-code platform that allows organizations to build and customize applications rapidly. Many processes can be configured using graphical interfaces and predefined platform components, reducing the need for extensive programming. However, the platform also provides scripting capabilities for implementing more complex business logic and extending standard functionality.

JavaScript is the primary scripting language used within ServiceNow. It can be used to implement server-side logic, control client-side behavior, validate data, and automate specific operations according to business requirements. This combination of configuration and scripting allows developers to adapt the platform while maintaining a flexible and scalable development environment.

For the Service Request Fulfillment project, ServiceNow configuration was mainly used to create and customize Service Catalog items, define request variables, control form behavior, configure approval and fulfillment processes, establish assignment rules and notifications, and implement Service Level Agreements (SLAs). These elements provided the technical foundation for the automation of the request fulfillment processes.

2.3 Information Technology Service Management (ITSM)

2.3.1 ITSM Overview

Information Technology Service Management (ITSM) is a strategic approach to designing, delivering, managing, and improving the way information technology services are provided within an organization. The primary objective of ITSM is to ensure that the appropriate processes, people, and technologies are in place to support the organization in achieving its business goals. This approach shifts the focus from managing isolated IT infrastructure components to delivering integrated and effective IT services.

2.3.2 ITIL and Service Management

ITIL (Information Technology Infrastructure Library) is a widely adopted framework that provides guidance and best practices for IT Service Management. It helps organizations establish structured approaches for designing, delivering, supporting, and continually improving IT-enabled services.

ITIL emphasizes the importance of defining clear processes, responsibilities, and service objectives. Its practices can be adapted to the specific requirements of an organization rather than being implemented as a rigid set of procedures.

For this project, ITIL concepts provide a general framework for understanding service request management and the importance of standardized processes, defined approval procedures, controlled fulfillment, and service-level monitoring.

2.3.3 Service Request Management

Service Request Management is an ITSM practice focused on handling standard requests submitted by users for predefined IT services. These requests may include application access, software installation, hardware requests, or password resets.

A structured request management process ensures that requests are properly recorded, validated, authorized, assigned, fulfilled, and closed. This improves the consistency and traceability of service delivery while ensuring that requests follow predefined procedures.

In ServiceNow, Service Request Management is supported by the **Service Catalog**, which provides users with a structured interface for submitting requests. Each catalog

item defines the required information and can be associated with approval processes, fulfillment tasks, assignment rules, notifications, and automation.

The process can therefore be summarized through several main activities: request submission, approval, assignment, fulfillment, and closure. In the context of this project, five types of IT service requests were considered:

1. **User Account Creation**
2. **Enterprise VPN Access**
3. **Software Installation**
4. **IT Hardware Requests**
5. **Password Reset**

These request types form the basis of the Service Request Fulfillment solution developed during the internship.

2.3.4 Request Fulfillment Lifecycle

The fulfillment of an IT service request follows a structured lifecycle that ensures the request is properly processed from submission to completion. A typical request fulfillment process consists of several stages, including submission, approval, assignment, fulfillment, and closure.

1. **Request Submission:** The user selects the required service and provides the necessary information through the Service Catalog.
2. **Approval:** The request is reviewed and, when required, submitted to the appropriate person or department for approval.
3. **Assignment:** Once approved, the request is assigned to the appropriate fulfillment group or generates the required fulfillment tasks.
4. **Fulfillment:** The responsible team performs the activities required to provide the requested service.

5. **Closure:** After the required activities are completed, the request is validated and closed.

In ServiceNow, these stages can be supported and automated through workflows, approvals, task assignment, notifications, and other platform capabilities. This automation helps reduce manual processing, improve traceability, and ensure that requests follow a consistent fulfillment process.

Figure 2.4 illustrates the main stages of the request fulfillment lifecycle.

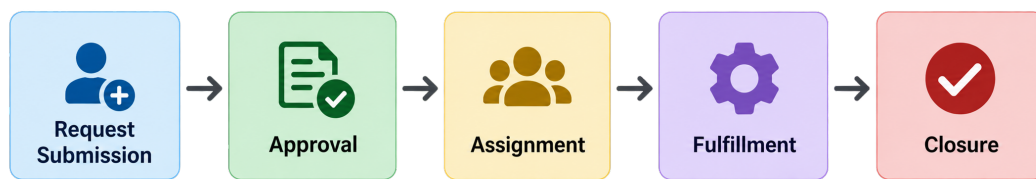


Figure 2.4: Request Fulfillment Lifecycle in ServiceNow

2.4 ServiceNow Service Catalog

2.4.1 Service Catalog and Catalog Items

The Service Catalog is the user-facing interface of the ITSM module, providing a centralized portal where employees can browse, search for, and request predefined IT services. It standardizes the submission of service requests by providing structured forms and ensuring that the necessary information is collected from the beginning of the request process.

Catalog Items represent the individual services available for request within the Service Catalog. Each item is designed to address a specific user need and can be organized into appropriate categories to facilitate navigation and selection.

For this project, five Catalog Items were designed to address common IT service needs: **User Account Creation**, **Enterprise VPN Access**, **Software Installation**, **IT Hardware Requests**, and **Password Reset**. Each item was configured with the information and process requirements necessary for its subsequent fulfillment.

2.4.2 Catalog Configuration and Access Control

ServiceNow provides several configuration features that allow Catalog Items to be adapted to the specific requirements of each service. One of the main elements is the use of **variables**, which define the information that users must provide when submitting a request. Variables can be used to collect different types of information, such as text, selections, dates, or references to existing records.

Variable Sets can also be used to group related variables and reuse them across multiple Catalog Items. This helps maintain consistency and reduces the need to configure the same information repeatedly.

The behavior and visibility of form fields can be controlled using **UI Policies**. These policies allow fields to be made mandatory, visible, or read-only depending on the conditions defined for a request. This makes it possible to create dynamic forms that display only the information relevant to the selected service.

Access to Catalog Items can be managed using **User Criteria**, which define which users or groups are authorized to access specific services. This ensures that catalog services are presented only to the appropriate users according to the organization's access requirements.

For the Service Request Fulfillment project, these configuration mechanisms were used to structure the request forms, collect the required information, control form behavior, and manage access to the different Catalog Items.

2.4.3 Catalog Configuration and Access Control

To collect the information required for a specific request, Catalog Items use **Variables** and **Variable Sets**, which act as input fields on the request form. These elements allow each Catalog Item to collect the information needed for its specific fulfillment process.

To provide a consistent user experience, **UI Policies** and **Client Scripts** can be used to dynamically control form fields, making them visible, mandatory, or read-only according to predefined conditions. In addition, **User Criteria** can be used to control access to Catalog Items and ensure that services are available only to authorized users.

For the Service Request Fulfillment project, these configuration features were used to structure the request forms, collect the required information, control form behavior, and

manage access to the different services.

2.5 Workflow Automation

2.5.1 Flow Designer and Process Automation

Flow Designer is ServiceNow's visual automation tool for creating multi-step processes. It allows developers to define a sequence of actions that are automatically executed when a predefined trigger occurs. These actions can include creating tasks, requesting approvals, updating records, and sending notifications.

Flow Designer helps replace manual routing and processing by connecting the different stages of a service request into a structured workflow. In the Service Request Fulfillment project, it was used to automate the processing of requests and coordinate their approval and fulfillment activities.

ServiceNow also supports backend automation through **Business Rules**. These server-side scripts execute when records are inserted, updated, or queried, allowing additional business logic to be applied to the system. Together, Flow Designer and Business Rules provide mechanisms for automating both process-level and record-level operations.

2.5.2 Approvals, Tasks, and Notifications

Automated workflows rely on approvals, fulfillment tasks, and notifications to coordinate the different stages of a request.

- **Approvals:** Approval requests are generated when a service requires authorization before fulfillment can continue. The approval process can involve managers or other responsible teams depending on the type of request.
- **Catalog Tasks (SCTASKs):** Once the required approvals are completed, fulfillment tasks can be generated and assigned to the appropriate support groups. These tasks represent the activities required to complete the requested service.
- **Notifications:** Automated notifications keep users and stakeholders informed about important events, such as pending approvals, task assignments, status changes, and request completion.

These mechanisms work together to provide a structured and traceable fulfillment process.

2.6 Service Level Agreements

2.6.1 SLA Concepts

A **Service Level Agreement (SLA)** is a documented commitment that defines the expected level of service and measurable performance targets. SLAs commonly specify response or fulfillment times and provide a basis for monitoring service performance.

In IT Service Management, SLAs help establish clear expectations, measure performance, and identify requests that are approaching or exceeding their defined service targets.

2.6.2 SLA Management in ServiceNow

In ServiceNow, **SLA Definitions** establish the conditions used to track the time associated with a record. These definitions can specify when an SLA starts, pauses, resumes, and stops according to predefined conditions.

ServiceNow continuously tracks active SLAs and provides visibility into the remaining time and current status of each target. This allows support teams to monitor fulfillment performance and identify requests that may require attention before an SLA is breached.

For the Service Request Fulfillment project, SLAs were considered to monitor the time required to process service requests and to support compliance with defined fulfillment targets.

2.7 Development and Deployment

2.7.1 Development and Testing Environments

ServiceNow development commonly uses separate environments for development, testing, and production. This separation allows configurations to be developed and validated before they are made available to end users.

The main environments used in a typical ServiceNow development lifecycle are:

- **DEV:** The development environment where configurations, Catalog Items, forms, and workflows are created and modified.
- **TEST (UAT):** The testing environment used to validate the developed solution and verify that it meets the defined requirements.
- **PROD:** The production environment where the validated solution is made available to end users.

This separation contributes to a controlled development process and reduces the risk of introducing untested changes into the production environment.

2.7.2 Update Sets and Deployment

An **Update Set** is a ServiceNow mechanism used to capture and organize configuration changes made within an instance. It allows related customizations to be packaged and transferred between ServiceNow environments.

Update Sets can contain different configuration elements, such as Catalog Items, workflows, business rules, client scripts, and UI policies. They therefore facilitate the controlled migration of developed configurations from a development environment to a testing or production environment.

For this project, Update Sets were used to organize the developed configurations and prepare them for controlled migration between the different ServiceNow environments.

2.8 Conclusion

This chapter presented the main theoretical and technical concepts underlying the Service Request Fulfillment project. It introduced ServiceNow, ITSM, Service Request Management, the Service Catalog, workflow automation, SLAs, and the development and deployment process. These concepts provide the foundation for the following chapters, which focus on the design and practical implementation of the developed solution.

Chapter 3

Analysis and Design of the Service Request Fulfillment Solution

3.1 Introduction

This chapter bridges the gap between the theoretical concepts of ITSM and the practical design of the Service Request Fulfillment solution. It provides an analysis of the existing request management process, identifying its current limitations and defining the functional requirements for the proposed solution. The chapter then presents the solution design, including the proposed ServiceNow architecture, the structure of the Service Catalog, and the data model. Finally, it provides a detailed design specification for each of the five core Catalog Items, including their respective inputs, approval workflows, and Service Level Agreements (SLAs).

3.2 Analysis of the Existing Request Management Process

3.2.1 Existing Process and Current Challenges

Prior to the implementation of the proposed solution, IT service requests were managed through different communication channels, requiring IT support personnel to manually review, process, and route each request. This approach presented several operational challenges :

- **Data Inconsistency:** Without standardized request forms, submitted requests may lack important information required for their processing and fulfillment, re-

sulting in additional communication with the requester.

- **Manual Routing and Approval:** Requests and approval steps may require manual intervention to identify the appropriate responsible person or fulfillment group, which can increase processing time.
- **Lack of Visibility:** Without a centralized system for managing requests, users and support teams have limited visibility into the status and progress of ongoing requests.
- **Limited Performance Monitoring:** The absence of centralized request tracking makes it difficult to consistently monitor fulfillment times, evaluate service performance, and identify process bottlenecks.

These limitations highlighted the need for a centralized and automated solution capable of standardizing request submission, automating approval and routing processes, and improving the overall traceability of IT service requests.

3.2.2 Requirements Analysis

Based on the analysis of the existing request management process, a set of functional requirements was defined for the proposed Service Request Fulfillment solution. These requirements describe the main capabilities that the system should provide to standardize and automate the handling of IT service requests.

The solution should meet the following functional requirements:

- **Request Submission:** Provide users with a centralized Service Catalog through which they can select and submit predefined IT service requests.
- **Structured Data Collection:** Provide dedicated request forms containing the information required for each type of service.
- **Approval Management:** Support predefined approval processes for requests that require authorization before fulfillment.
- **Automatic Assignment:** Route requests and fulfillment tasks to the appropriate support groups according to predefined rules.

- **Workflow Automation:** Automate the different stages of the request fulfillment process, from submission and approval to task completion and closure.
- **Notifications:** Inform users and relevant stakeholders about important events and changes in the request lifecycle.
- **SLA Management:** Support the monitoring of fulfillment times according to predefined service-level targets.
- **Request Tracking:** Provide centralized visibility into the status and progress of submitted requests.

These requirements served as the basis for designing the proposed ServiceNow solution and defining the configuration of the different Catalog Items and fulfillment workflows.

3.3 Functional Analysis

3.3.1 Actors and Responsibilities

The Service Request Fulfillment process involves several actors who interact with the system at different stages of the request lifecycle. Each actor has specific responsibilities related to the submission, authorization, and fulfillment of service requests.

The main actors identified for the proposed solution are :

- **Requester:** The employee who submits an IT service request through the Service Catalog and provides the information required for its processing.
- **Direct Manager:** The responsible person who reviews and approves requests when managerial authorization is required.
- **Fulfillment Teams:** The teams responsible for processing assigned fulfillment tasks and providing the requested service.
- **HR Team:** The responsible team involved in requests that require employee-related validation, such as account creation.
- **Security Team:** The team responsible for reviewing and processing requests related to secure access and network services, such as Enterprise VPN Access.

The interaction between these actors depends on the type of service requested. A request may therefore involve one or more approval and fulfillment stages before it can be completed.

Figure 3.1 illustrates the main actors involved in the Service Request Fulfillment process and their interactions with the proposed solution.

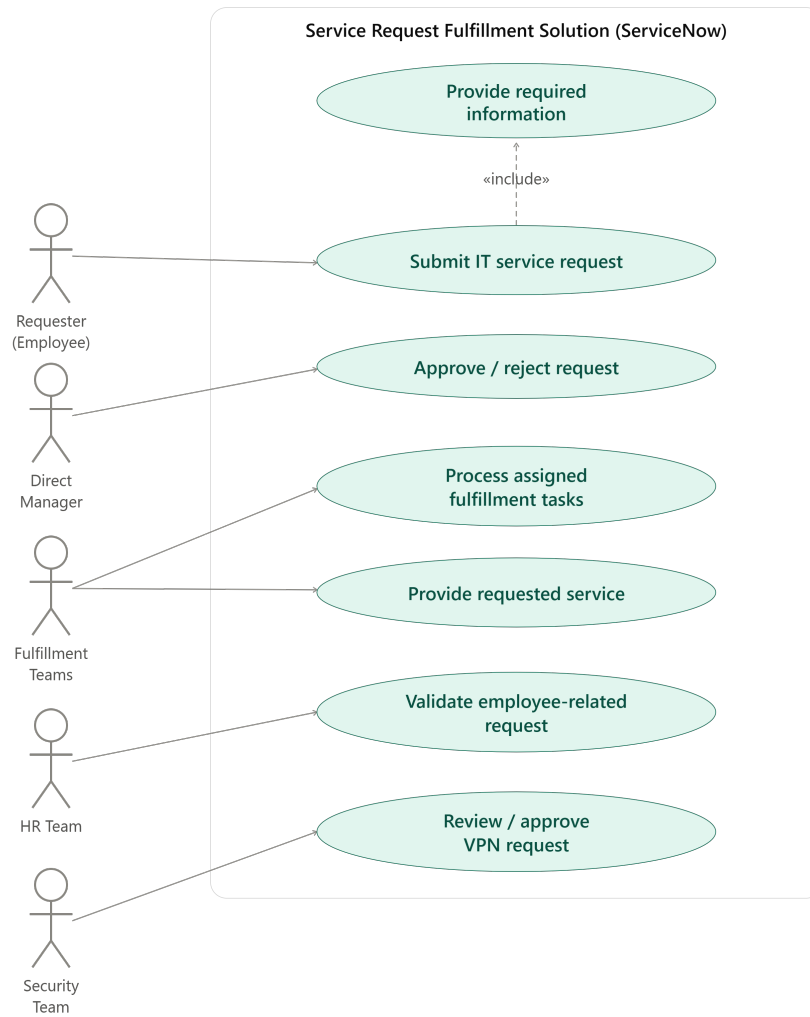


Figure 3.1: Use Case Diagram of the Service Request Fulfillment Solution

3.3.2 Service Request Requirements

Each Catalog Item requires specific information and fulfillment conditions depending on the type of service requested. The requirements were analyzed to determine the information to be collected, the approvals to be obtained, and the fulfillment activities to be performed for each request.

Catalog Item	Category	Main Requirements	Fulfillment Activities
Create User Account	User Management	User Concerned, Manager, Necessary Applications, Account Type, Start Date, Employee ID Number	AD account creation, mailbox creation, and application access
Enterprise VPN Access Request	Access Management	User Concerned, Access Type, Justification, Access Duration, Start Date, End Date	Security analysis, VPN configuration, User testing
Software Installation	Software Services	User Concerned, Software Requested, Computer, Version, Business Justification, Desired Date	License availability check, Software installation, User validation
IT Equipment Request	Hardware Services	User Concerned, Assigned Materials, Material Type, Reason, Cost Center, Quantity, Desired Date	Stock check, Equipment preparation, Delivery
Reset Password	Self Service	User Concerned, Application Concerned, Is Urgent, Reason	Password reset

Table 3.1: Main Requirements of the Catalog Items

The analysis of these requirements provided the basis for defining the specific forms, approval processes, conditions, and fulfillment tasks associated with each Catalog Item. The resulting workflows were designed to adapt the processing of each request according to its specific requirements.

3.3.3 Request and Approval Workflows

The analysis of the five Catalog Items showed that each request follows a specific fulfillment workflow based on its requirements and approval conditions. Although the workflows share common stages, such as request submission, approval, task fulfillment, and closure, each service includes specific conditions and activities.

The main workflow characteristics of the five Catalog Items are summarized in Table 3.2.

Catalog Item	Approval Process	Main Workflow Activities
Create User Account	Manager approval followed by HR approval	Determine account type and priority, create AD account, create mailbox, and assign application access when required.
Enterprise VPN Access Request	Manager approval followed by Security Team approval	Determine priority, perform security analysis, configure VPN, and conduct user testing.
Software Installation	Manager approval followed by user validation	Check license requirement and availability, install the software, obtain user validation, and close the request.
IT Equipment Request	Manager approval	Check equipment availability, determine priority, prepare and deliver the equipment, and update the CMDB.
Password Reset	No approval required	Determine priority according to urgency, create the password reset task, and close the request after completion.

Table 3.2: Main Characteristics of the Request Workflows

The functional analysis shows that the workflows cannot follow a single standard process. Instead, each workflow adapts to the characteristics and requirements of the requested service:

- **Simple Requests:** Requests such as Password Reset do not require managerial approval and can proceed directly to fulfillment, with their priority determined according to the urgency of the request.
- **Requests Requiring Managerial Approval:** Requests such as Software Installation and IT Equipment Request require Direct Manager approval before the fulfillment activities can begin.
- **Requests Requiring Multiple Approvals:** Requests such as User Account Creation and Enterprise VPN Access require additional validation after managerial approval, involving the HR Team or Security Team respectively.

The workflows also include conditional paths to handle specific situations. For example, rejected approvals result in the request being closed as incomplete, while conditions such as account type, software license availability, equipment stock, and request urgency can modify the subsequent processing of a request.

This approach allows each Catalog Item to follow a controlled process adapted to its specific requirements while maintaining a consistent overall fulfillment structure.

3.4 Solution Design

3.4.1 Proposed ServiceNow Architecture

The proposed solution is built on the ServiceNow ITSM architecture and uses the platform's request management structure to organize the fulfillment of IT service requests. When a user submits a Catalog Item, ServiceNow creates a hierarchy of related records that supports the request throughout its lifecycle.

1. **Request (REQ):** Represents the overall request submitted by the user and acts as the container for the requested services.
2. **Requested Item (RITM):** Represents the specific Catalog Item requested by the user and contains the information collected through its request form.
3. **Catalog Task (SCTASK):** Represents an individual fulfillment activity generated for the Requested Item and assigned to the appropriate fulfillment team.

The RITM serves as the central element of the fulfillment process. Once it is created, the corresponding Flow Designer workflow manages the required conditions, approvals, and fulfillment tasks. Depending on the requested service, one or more SCTASKs can be generated and assigned to the appropriate teams until the request is completed.

This architecture provides a structured relationship between the user request, the requested service, and the activities required for its fulfillment, while supporting centralized tracking of the request lifecycle.

Figure 3.2 illustrates the proposed architecture of the Service Request Fulfillment solution.

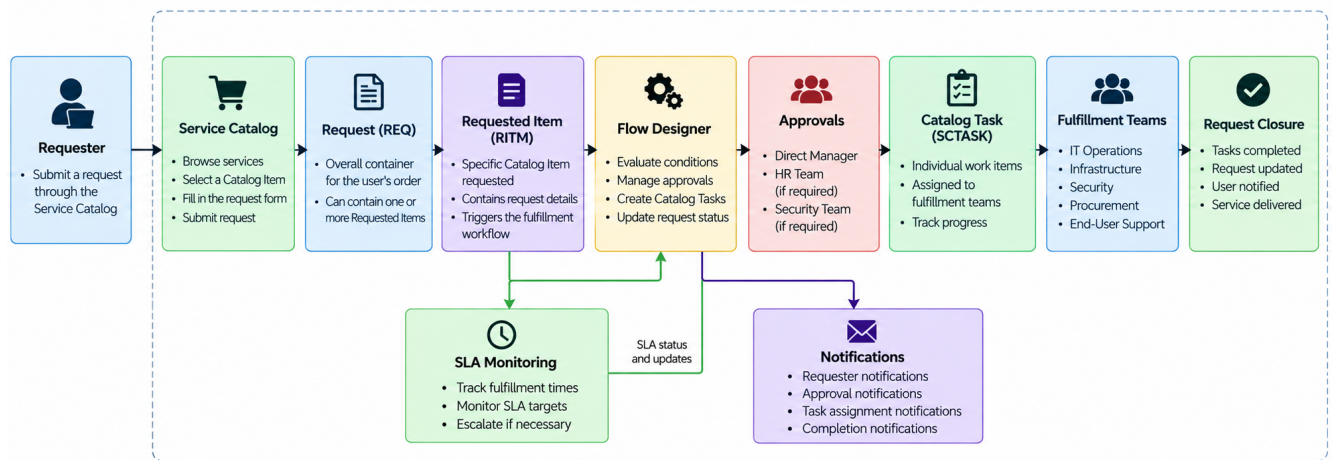


Figure 3.2: Proposed Architecture of the Service Request Fulfillment Solution

3.4.2 Service Catalog Structure

The proposed Service Catalog is organized around five Catalog Items, each corresponding to a specific type of IT service request. This structure allows users to identify the required service easily while ensuring that each request is associated with the appropriate information and fulfillment process.

The Catalog Items are organized into logical categories according to the type of service they provide:

- **User Management:** Create User Account
- **Access Management:** Enterprise VPN Access Request
- **Software Services:** Software Installation

- **Hardware Services:** IT Equipment Request
- **Self Service:** Reset Password

Each Catalog Item is associated with a dedicated request form containing the variables required for its processing. The information collected through these forms is then used by the corresponding workflow to determine the required conditions, approvals, and fulfillment activities.

Figure 3.3 illustrates the organization of the proposed Service Catalog.

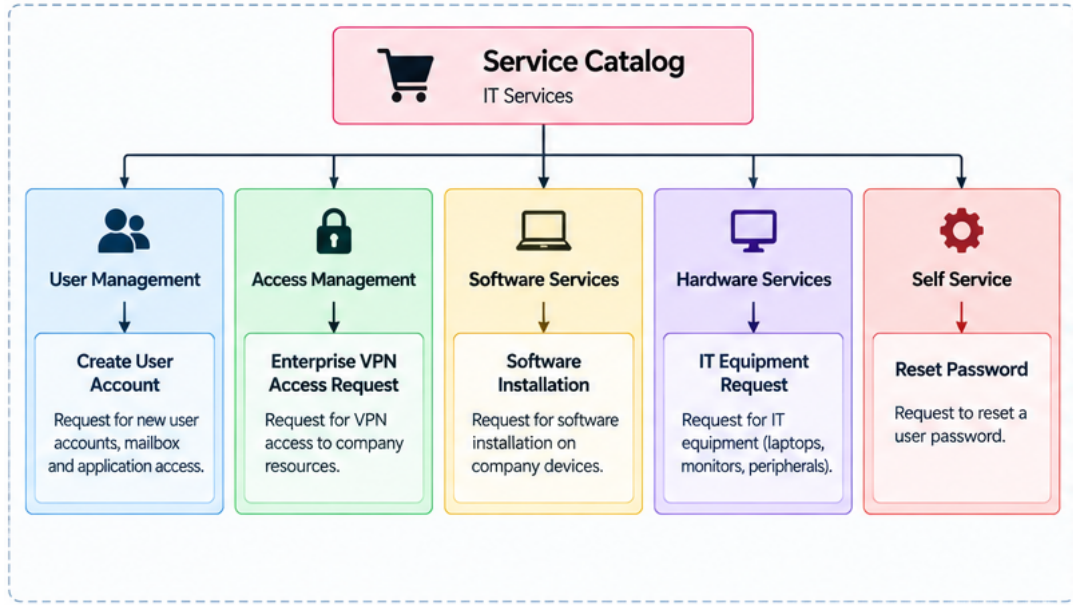


Figure 3.3: Structure of the Service Catalog

3.4.3 Request Fulfillment Workflow

The proposed fulfillment process follows a common structure while allowing each Catalog Item to include specific conditions and activities. Once a request is submitted, the corresponding Requested Item is processed by the associated workflow.

The general workflow consists of the following stages:

1. **Request Submission:** The user selects a Catalog Item and provides the required information.
2. **Initial Processing:** The workflow retrieves the submitted variables and evaluates conditions that may affect the request, such as account type, urgency, license requirements, or equipment availability.

3. **Approval:** When required, the request is submitted to the appropriate approver. Depending on the service, this may involve the Direct Manager, HR Team, or Security Team.
4. **Rejection Handling:** If an approval is rejected, the workflow updates the request status and closes the Requested Item as incomplete.
5. **Fulfillment:** For approved requests, the workflow creates the required Catalog Tasks and assigns them to the appropriate fulfillment teams.
6. **Validation and Closure:** Where required, the completed service is validated before the Requested Item is marked as complete.

Although this general structure is shared across the solution, the workflow branches according to the requirements of each service. For example, the User Account Creation workflow includes Manager and HR approval, while the Enterprise VPN Access workflow includes Manager and Security approval. Software Installation includes a license check and user validation, whereas IT Equipment Request includes stock verification and CMDB updating. Password Reset follows a simpler process based primarily on request urgency.

Figure 3.4 presents the general workflow designed for the Service Request Fulfillment solution.

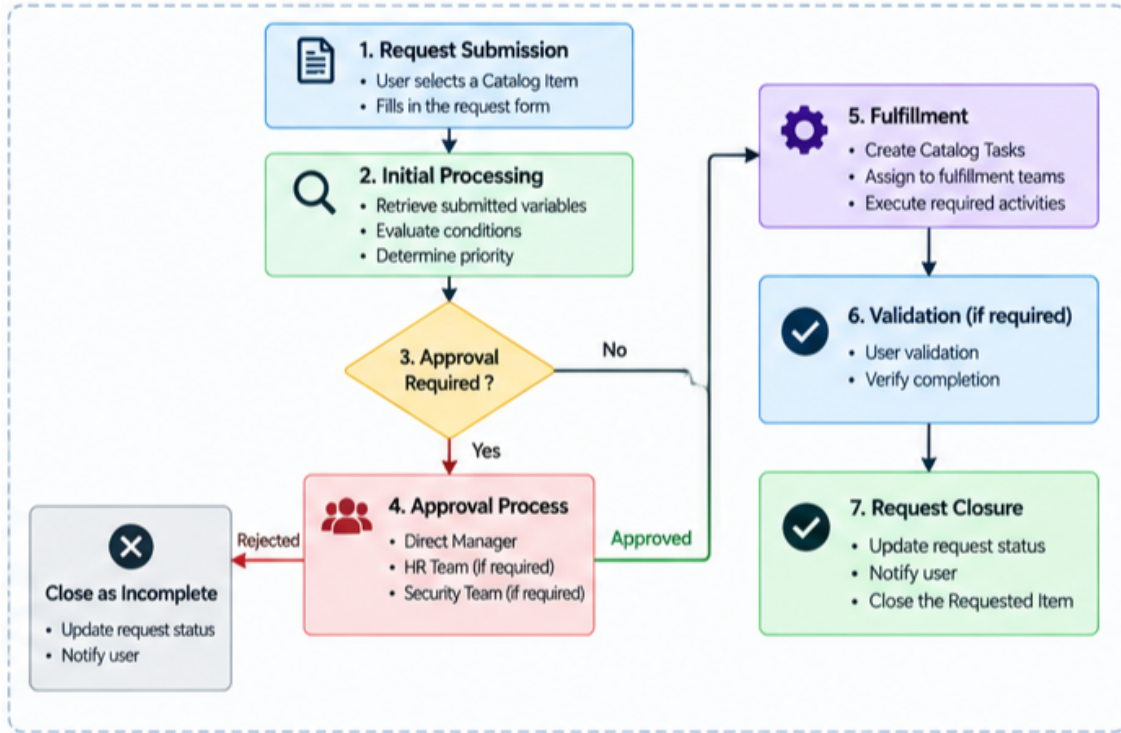


Figure 3.4: General Request Fulfillment Workflow

3.4.4 Request Data and Configuration Model

The proposed solution combines ServiceNow configuration elements with the standard request records generated during the fulfillment process. This structure ensures that the information provided by the requester is preserved and made available throughout the different stages of request processing.

The main elements involved are:

- **Catalog Item:** Defines the service available to users and the information required to submit the request.
- **Variables:** Collect the information provided by the requester through the Catalog Item form. These values are used by the workflow to evaluate conditions and determine the required processing.
- **Request (REQ):** Represents the overall request submitted by the user and serves as a container for one or more Requested Items.

- **Requested Item (RITM):** Represents the specific Catalog Item requested and serves as the main record throughout its fulfillment lifecycle.
- **Approval Records:** Store the approval requests and decisions associated with the Requested Item when authorization is required.
- **Catalog Tasks (SCTASKs):** Represent the individual fulfillment activities generated from the Requested Item and assigned to the appropriate fulfillment teams.

The configuration elements supporting this structure include Flow Designer, UI Policies, User Criteria, Business Rules, Service Level Agreements, and Notifications. These elements interact with the request records to control processing, apply conditions, manage access, monitor fulfillment times, and communicate important updates.

Figure 3.5 illustrates the relationships between the main configuration elements and request records used in the proposed solution.

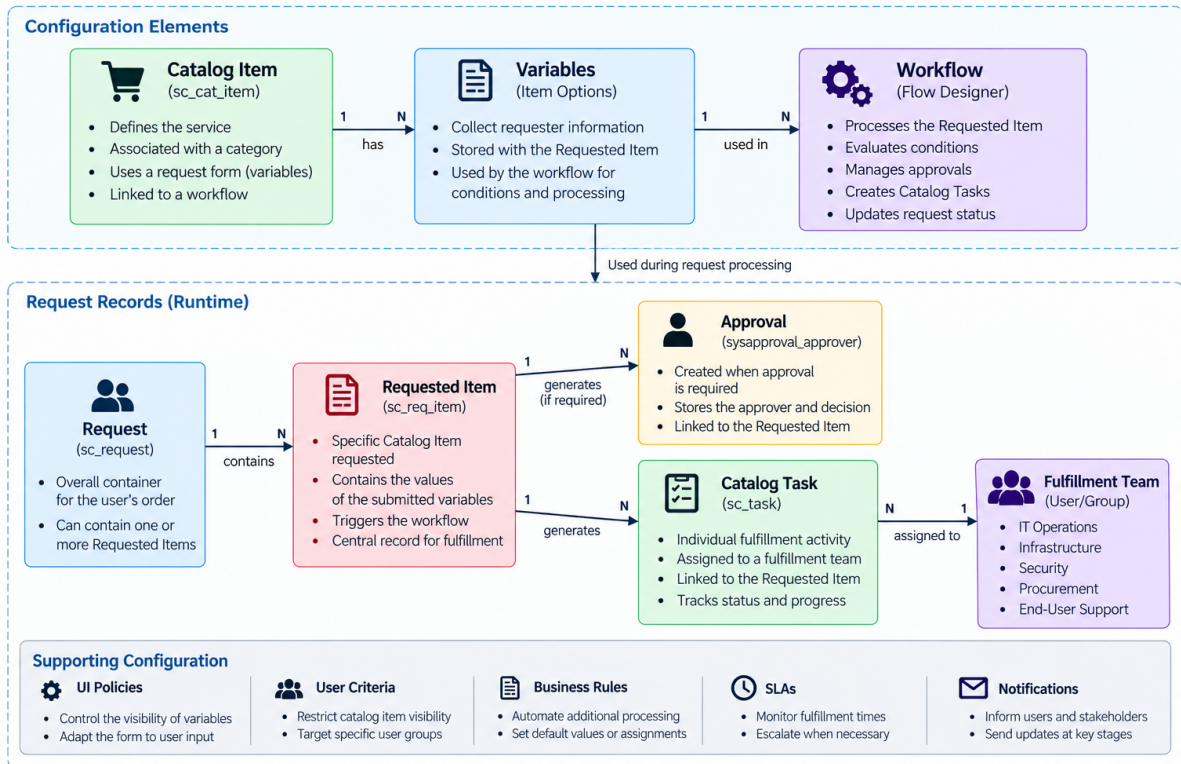


Figure 3.5: Data and Configuration Model of the Solution

3.5 Design of the Catalog Items

3.5.1 Create User Account

The Create User Account Catalog Item is designed to manage requests for the creation of a user account and the resources required for a new employee. The request form collects the information necessary to identify the user, determine the account type, and specify any additional applications required.

The main design elements are presented in Table 3.3.

Design Element	Description
Required Inputs	User Concerned, Manager, Necessary Applications, Account Type, Start Date, and Employee ID Number
Approval Process	Direct Manager approval followed by HR approval
Conditions	The account type is evaluated to determine the appropriate priority, while the requested applications determine whether additional application access is required.
Fulfillment	AD account creation, mailbox creation, and application access when required
Completion	The Requested Item is completed after the required fulfillment activities have been successfully processed.

Table 3.3: Design of the Create User Account Catalog Item

The workflow follows a sequential approval process. The request is first submitted for Direct Manager approval and, after successful approval, for HR approval. If an approval is rejected, the request follows the incomplete closure path. When the required approvals are obtained, the corresponding fulfillment tasks are generated. The workflow also includes conditional processing based on the account type and the applications requested.

Figure 3.6 presents the designed fulfillment workflow for the Create User Account Catalog Item.

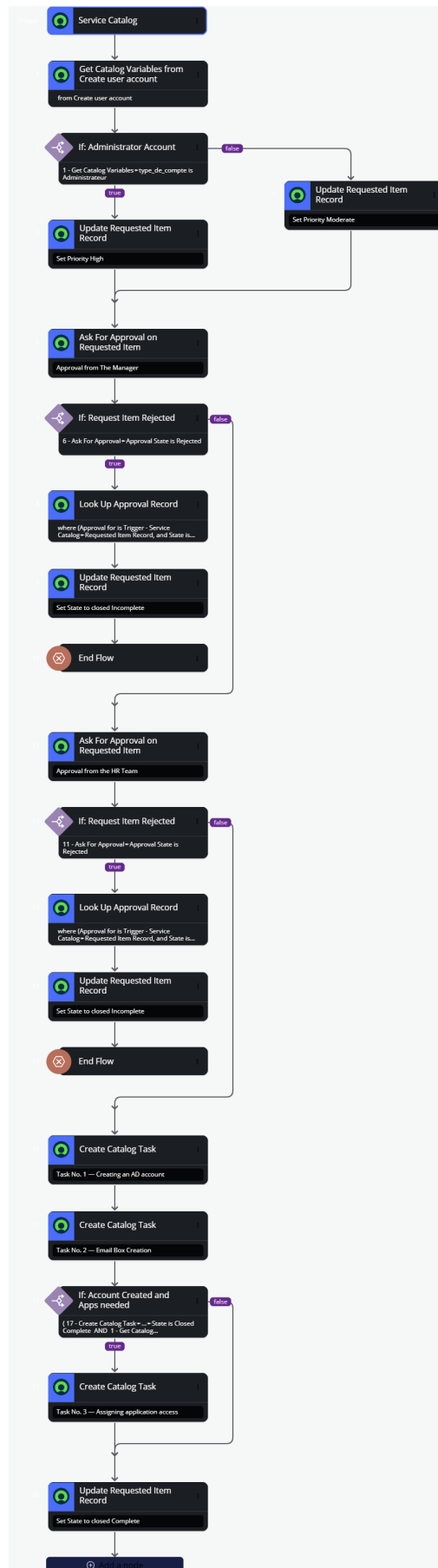


Figure 3.6: Create User Account Fulfillment Workflow

3.5.2 Enterprise VPN Access Request

The Enterprise VPN Access Request Catalog Item is designed to manage requests for secure remote access to enterprise resources. Due to the nature of this service. Due to the nature of this service, the request requires both managerial authorization and security validation before the technical fulfillment activities can begin.

The main design elements are presented in Table 3.4.

Design Element	Description
Required Inputs	User Concerned, Access Type, Justification, Access Duration, Start Date, and End Date
Approval Process	Direct Manager approval followed by Security Team approval
Conditions	The request priority is determined according to the defined conditions. Rejection at either approval stage stops the fulfillment process.
Fulfillment	Security analysis, VPN configuration, and user testing
Completion	The Requested Item is completed after the required fulfillment activities and user testing have been completed.

Table 3.4: Design of the Enterprise VPN Access Request Catalog Item

The workflow implements two sequential authorization stages. The Direct Manager first reviews the request, followed by the Security Team. If either approval is rejected, the request follows the incomplete closure path. Otherwise, the workflow proceeds to the security analysis, VPN configuration, and user testing activities.

Figure 3.7 presents the designed fulfillment workflow for the Enterprise VPN Access Request Catalog Item.

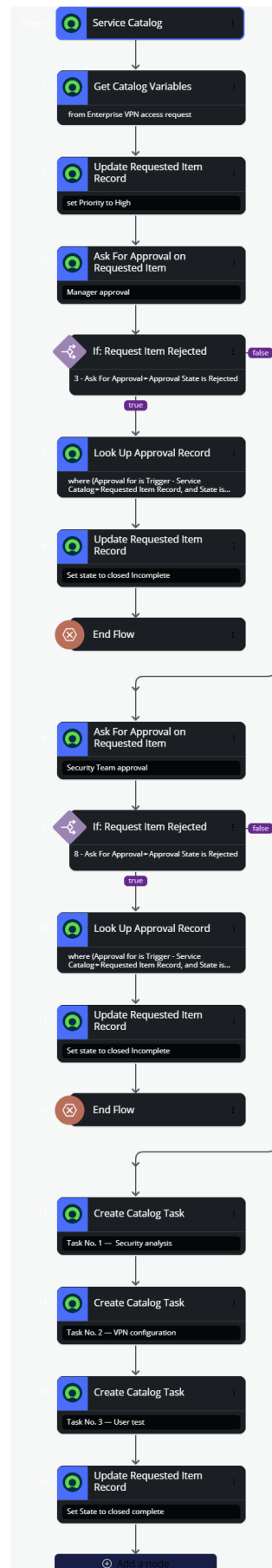


Figure 3.7: Enterprise VPN Access Request Fulfillment Workflow

3.5.3 Software Installation

The Software Installation Catalog Item is designed to manage requests for installing software on a user's computer. The request form collects the information required to identify the requested software and the target computer. The workflow also includes a condition to determine whether a license check is required.

The main design elements are presented in Table 3.5.

Design Element	Description
Required Inputs	User Concerned, Software Requested, Computer, Version, Business Justification, and Desired Date
Approval Process	Direct Manager approval followed by user validation after installation
Conditions	The workflow evaluates whether a license is required and, when applicable, checks its availability before installation.
Fulfillment	License availability check and software installation
Completion	The user validates the installation before the Requested Item is completed.

Table 3.5: Design of the Software Installation Catalog Item

After submission, the workflow evaluates the license requirement and processes the request through the managerial approval stage. Once approved, the required fulfillment activities are performed. The user is then asked to validate the installation before the request is completed.

Figure 3.8 presents the designed fulfillment workflow for the Software Installation Catalog Item.

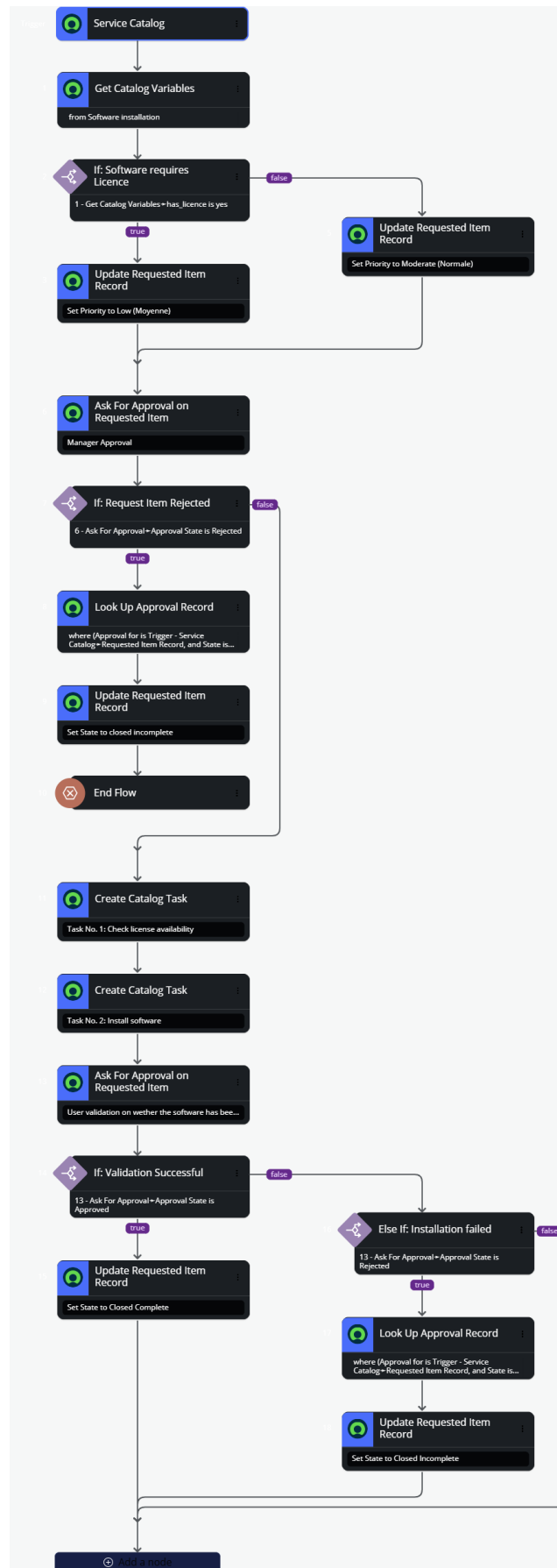


Figure 3.8: Software Installation Fulfillment Workflow

3.5.4 IT Equipment Request

The IT Equipment Request Catalog Item is designed to manage requests for IT hardware and equipment. The process combines managerial approval with an equipment availability check before the required fulfillment activities are performed.

The main design elements are presented in Table 3.6.

Design Element	Description
Required Inputs	User Concerned, Assigned Materials, Material Type, Reason, Cost Center, Quantity, and Desired Date
Approval Process	Direct Manager approval
Conditions	The workflow checks equipment availability and determines the subsequent processing according to the stock condition.
Fulfillment	Stock verification, equipment preparation, and equipment delivery
Additional Processing	The corresponding equipment information is updated in the CMDB as part of the fulfillment process.
Completion	The Requested Item is completed after the required fulfillment activities have been processed.

Table 3.6: Design of the IT Equipment Request Catalog Item

Following managerial approval, the workflow performs a stock verification. The subsequent processing depends on equipment availability. The fulfillment process then includes equipment preparation and delivery, followed by the required CMDB update before the request is completed.

Figure 3.9 presents the designed fulfillment workflow for the IT Equipment Request Catalog Item.

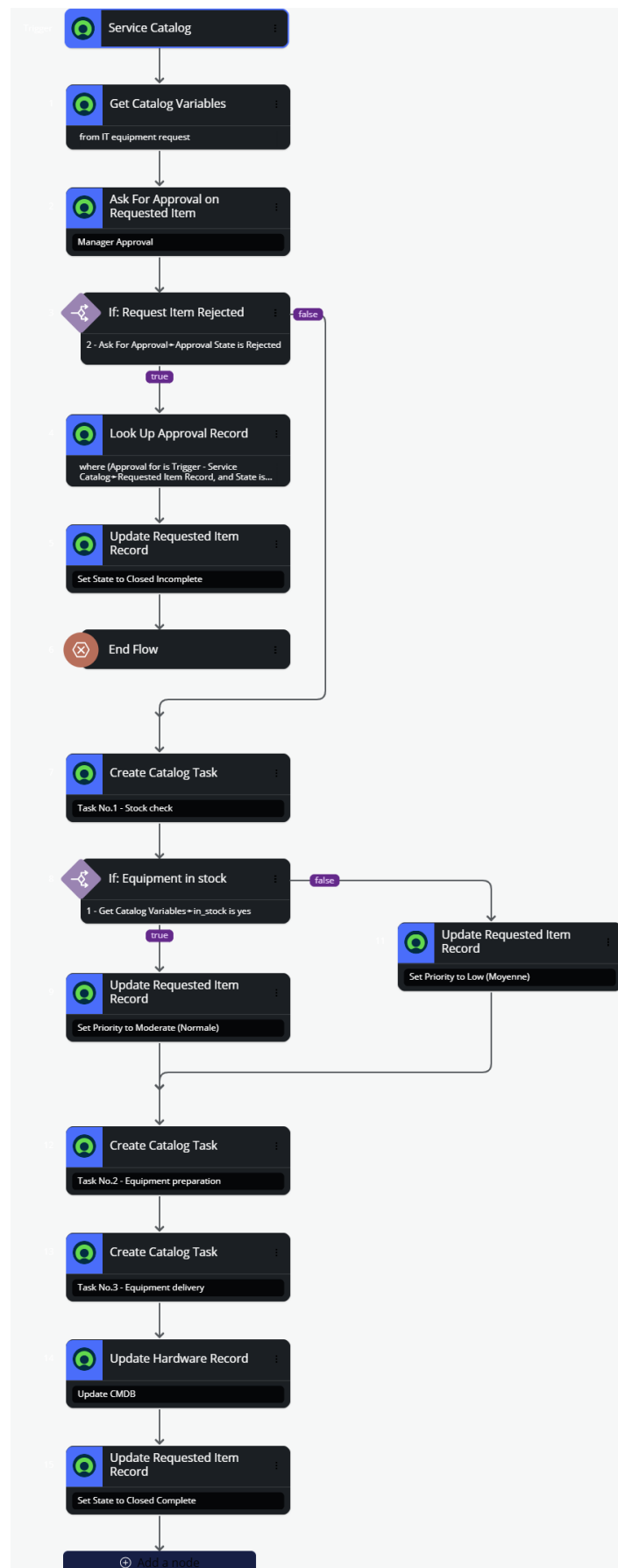


Figure 3.9: IT Equipment Request Fulfillment Workflow

3.5.5 Reset Password

The **Reset Password** Catalog Item was designed to provide users with a simple way to request a password reset. Unlike the other Catalog Items, this request does not require managerial approval. Its processing is mainly based on the urgency of the request and the execution of a fulfillment task.

Design Element	Description
Required Inputs	User Concerned, Application Concerned, Urgency, Reason
Approval	No approval required
Conditions	The urgency of the request determines its priority
Fulfillment	Creation of a password reset Catalog Task assigned to the Service Desk
Completion	The Requested Item is closed after the fulfillment task is completed

Table 3.7: Design of the Reset Password Catalog Item

The workflow was designed to evaluate the urgency of the request, create a fulfillment task for the Service Desk, and close the Requested Item once the task is completed. This provides a lightweight process suitable for a self-service password reset request.

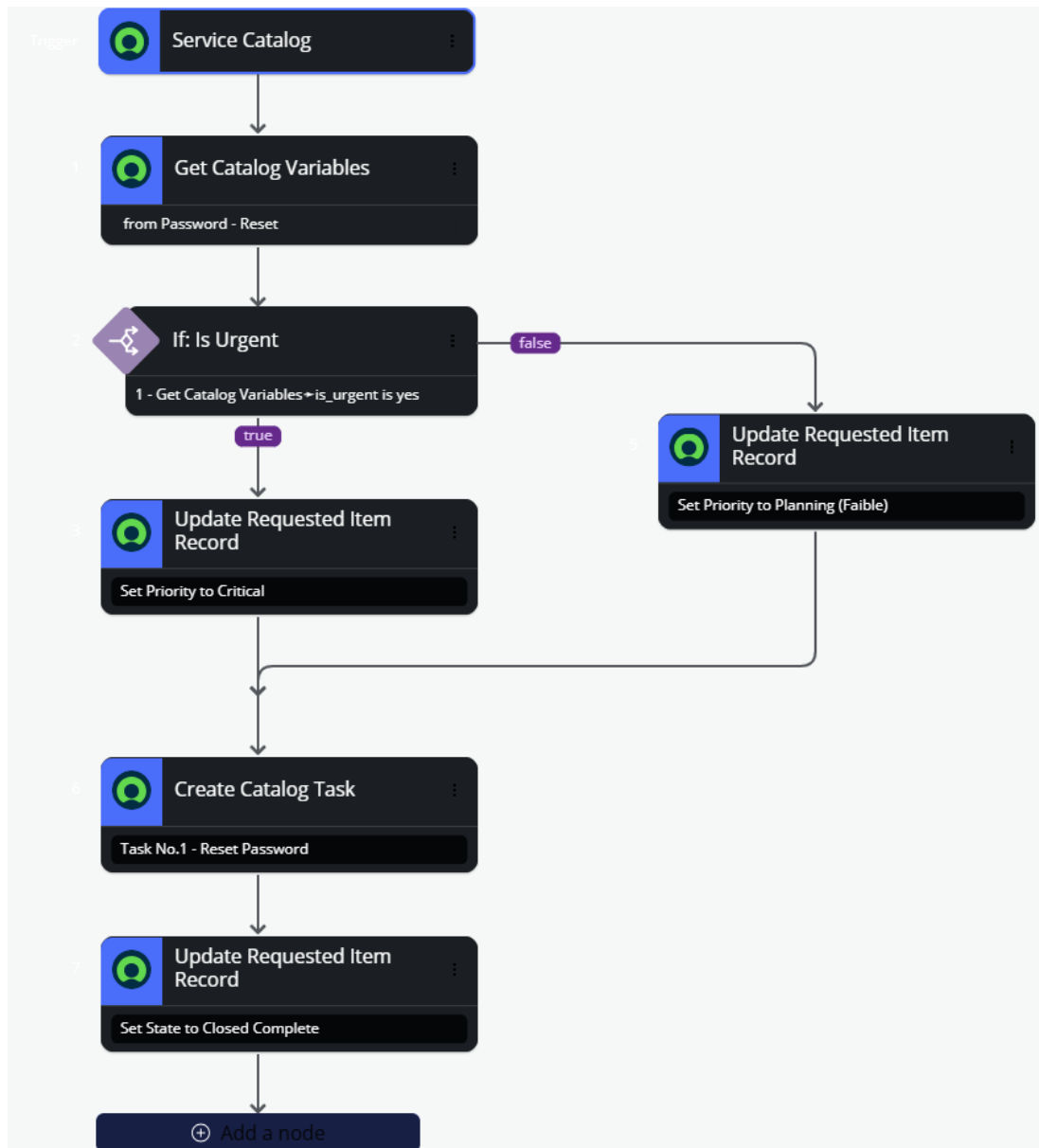


Figure 3.10: Reset Password Fulfillment Workflow

Alternative Business Rule Implementation As an additional technical approach, Business Rules were also developed as a possible alternative to the Flow Designer implementation. The objective was to reproduce the main processing logic of the Reset Password request using server-side JavaScript executed directly on the relevant ServiceNow records.

Three Business Rules were created for this purpose:

- **BR - Password Reset - Set Priority:** determines the priority of the Requested Item according to the urgency selected by the requester. An urgent request is

assigned the highest priority, while a non-urgent request receives a lower priority.

- **BR - Password Reset - Create Catalog Task:** automatically creates a Catalog Task for the Service Desk when a Reset Password request is submitted. The task contains information about the requested user and the concerned application.
- **BR - Password Reset - Close RITM:** monitors the completion of the Catalog Task and automatically updates the associated Requested Item to a closed state once the task is completed.

This alternative implementation demonstrates that the same fulfillment logic can be achieved through server-side Business Rules in addition to Flow Designer. The Business Rules therefore provide a possible code-based alternative for implementing the workflow logic, while the Flow Designer implementation offers a visual and more easily maintainable representation of the process.

3.5.6 Service Level Agreement Design

Service Level Agreements (SLAs) were defined to establish the expected fulfillment time for each type of service request. The applicable SLA depends on the Catalog Item and, where applicable, the priority assigned to the request. This allows requests with different levels of urgency or complexity to be associated with appropriate fulfillment targets.

The configured SLA targets for the five Catalog Items are presented in Table 3.8.

Catalog Item	Priority	SLA Target	SLA Definition
Create User Account	Normal	24 hours	Standard user account creation
Create User Account	High	48 hours	Administrator account creation
Enterprise VPN Access Request	High	4 to 24 hours	Enterprise VPN access request
Software Installation	Normal	1 business day	Standard software installation
Software Installation	Medium	3 business days	Licensed software installation
IT Equipment Request	Normal	2 business days	Request for in-stock equipment
IT Equipment Request	Medium	10 business days	Request for out-of-stock equipment
Reset Password	Low	2 hours	Standard password reset
Reset Password	Critical	30 minutes	Urgent password reset

Table 3.8: SLA Targets Defined for the Service Catalog Items

The SLA definitions establish a relationship between request priority and expected fulfillment time. For example, urgent password resets have a Critical priority and a 30-minute target, while standard resets have a Low priority and a 2-hour target.

These SLA requirements complement the fulfillment workflows and are configured in ServiceNow according to the defined conditions.

3.6 Conclusion

This chapter translated existing request management challenges into functional requirements to design the new Service Request Fulfillment solution. The proposed ServiceNow

architecture detailed the Service Catalog, workflows, priority-based SLAs, and five core Catalog Items including a Business Rule approach for Password Resets. The following chapter covers the system's practical implementation and configuration.

Chapter 4

Implementation of the Service Request Fulfillment Solution

4.1 Introduction

This chapter details the practical implementation of the Service Request Fulfillment solution designed in the previous phase, providing a comprehensive overview of the ServiceNow development environment and its core configurations. It explores the technical realization of the Service Catalog, automated workflows, specific Catalog Items, Service Level Agreements (SLAs), and supporting automation scripts. Finally, the chapter outlines the rigorous testing and validation activities conducted to ensure system reliability, culminating in the packaging and preparation of these configurations for seamless deployment across different ServiceNow instances.

4.2 ServiceNow Development Environment

4.2.1 Development Instance

The implementation was carried out using a Personal Developer Instance (PDI) provided by ServiceNow. The PDI, identified as **dev292675**, was used as the development environment for configuring and testing the Service Request Fulfillment solution.

The instance was used in the context of ServiceNow System Administrator training and provided the environment required to apply the concepts learned during the training to the practical project. It was running the **Australia** release and was used to develop

and test the Catalog Items, workflows, Business Rules, SLAs, and other configuration elements of the solution.

The PDI therefore served as the main development and testing environment throughout the practical phase of the internship.

4.2.2 Application and Modules Used

The implementation was mainly carried out within the **IT Service Management (ITSM)** environment of ServiceNow. Several applications and modules were used throughout the development of the Service Request Fulfillment solution.

The main components used were:

- **Service Catalog and Catalog Builder:** used to create and configure the five Catalog Items, their categories, request forms, and variables.
- **Flow Designer / Process Automation:** used to implement request workflows, including conditions, approvals, fulfillment tasks, and notifications.
- **SLA Definitions:** used to configure service-level targets according to the type and priority of each request.
- **System Definition:** used to configure supporting elements such as Business Rules and other system-level components.
- **User and Group Management:** used to manage the users and groups involved in request submission, approval, and task assignment.

These components were combined to implement the different stages of the Service Request Fulfillment process, from request submission and approval to fulfillment and closure.

4.2.3 Configuration Environment

The solution was developed and configured directly within the ServiceNow PDI. The configuration environment provided access to the main components required for the implementation, including the Service Catalog, Flow Designer, Business Rules, SLAs, users, and groups.

The development activities were performed progressively, starting with the configuration of the Catalog Items and their variables, followed by the implementation of the fulfillment workflows and supporting automation. The configured components were then tested within the same development environment before preparing the solution for deployment.

4.3 Service Catalog Implementation

4.3.1 Catalog Creation and Organization

Before creating the Catalog Items, an IT catalog was created through **Maintain Catalogs** in the **Catalog Definition** module. This catalog served as the main container for the services developed as part of the project.

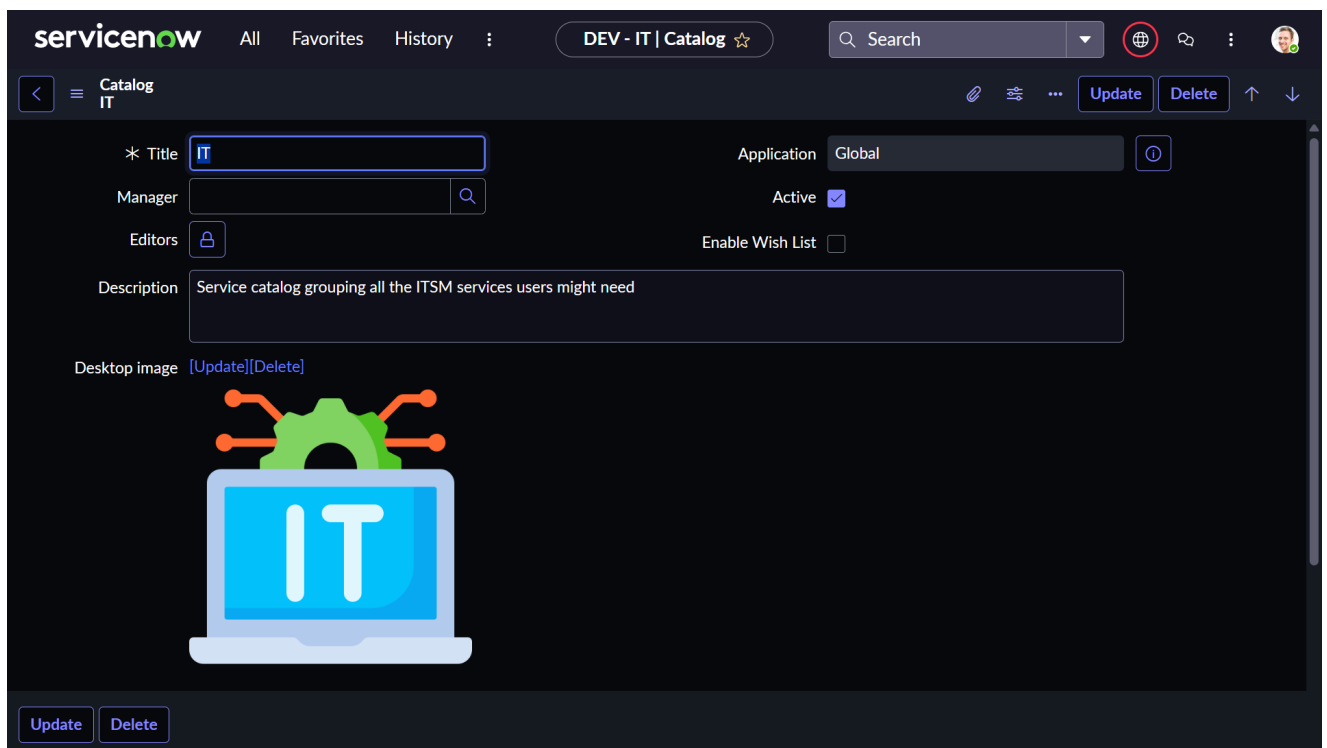


Figure 4.1: Creation of the IT Catalog

4.3.2 Catalog Item Creation

The Catalog Items were created using **Catalog Builder** within the **Service Catalog** module. Catalog Builder provided a structured interface for creating and configuring the services that would be made available to users.

Each Catalog Item was created according to the requirements identified during the analysis phase. The five services implemented in the project were:

- **Create User Account**
- **Enterprise VPN Access Request**
- **Software Installation**
- **IT Equipment Request**
- **Reset Password**

Figure 4.2 presents the Catalog Builder interface used during the implementation.

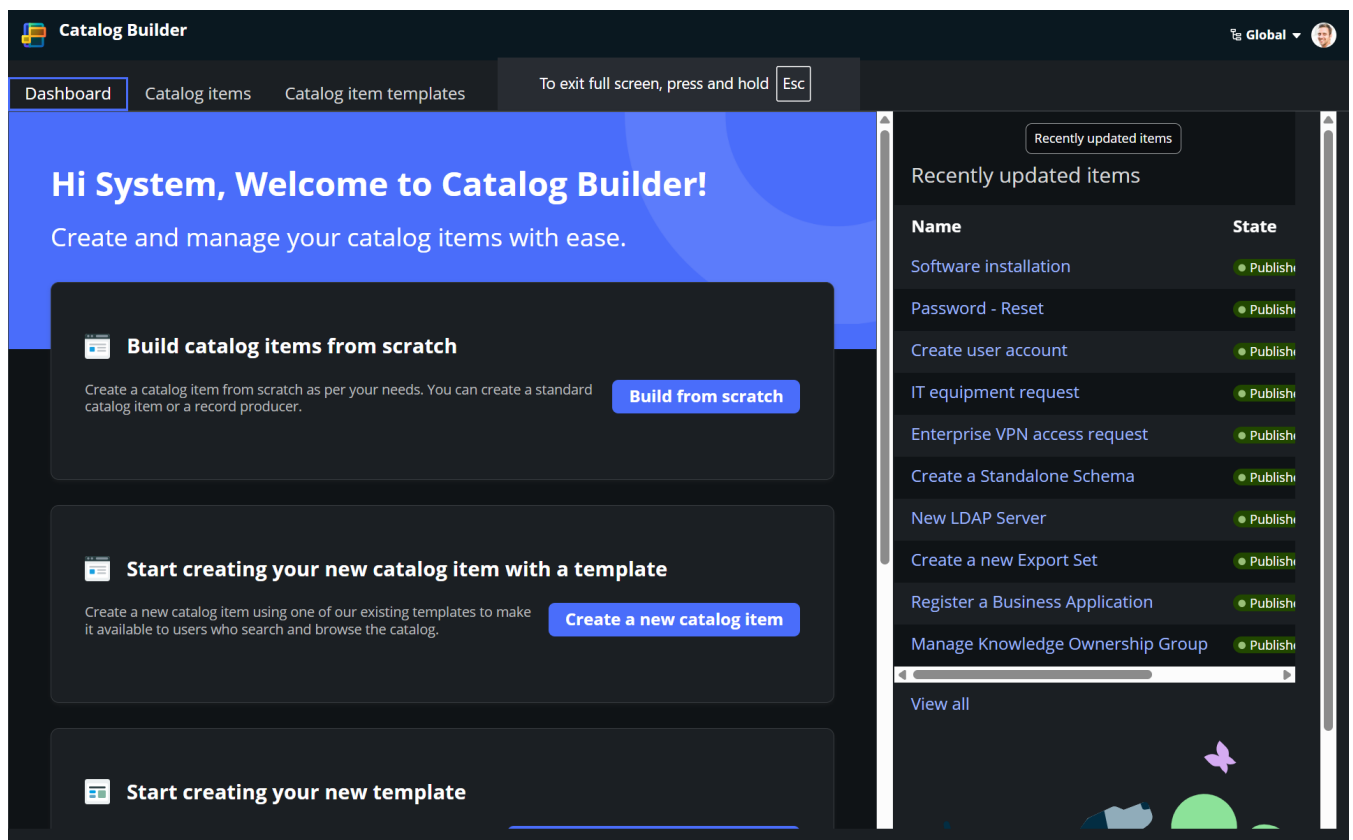


Figure 4.2: Catalog Builder

4.3.3 Variable and Request Form Configuration

After creating the Catalog Items, the corresponding request forms were configured using **Catalog Builder**. Variables were added to each form to collect the information required for processing the requested service.

The variables were configured according to the requirements defined during the analysis phase. Depending on the type of information, different field types were used, such as reference fields, choice lists, date fields, and text fields. Required information was also configured as mandatory where necessary.

The configuration of the variables and request forms allowed the information submitted by the requester to be structured and subsequently used by the fulfillment workflows.

Figure 4.3: Variable Configuration for a Catalog Item (Create User Account)

4.3.4 Categories and User Access Configuration

The Catalog Items were organized into dedicated categories according to the type of service provided. The categories used for the five Catalog Items are presented in Table 4.1.

Catalog Item	Category
Create User Account	User Management
Enterprise VPN Access Request	Access Management
Software Installation	Software Services
IT Equipment Request	Hardware Services
Reset Password	Self Service

Table 4.1: Catalog Items and Their Categories

Figure 4.4 illustrates the configuration of the catalog categories within the IT catalog.

The screenshot shows the ServiceNow interface for configuring a catalog category. The title is 'User Management' and the catalog is 'IT'. The application is set to 'Global' and the 'Active' checkbox is checked. The description is 'Groups all the User management tools and services'. There are also fields for 'Location' and 'Parent'. At the bottom, there are 'Update' and 'Delete' buttons.

Figure 4.4: Configuration of the Catalog Categories

Access to specific Catalog Items was controlled using **User Criteria**. These criteria determine which users can access and submit a particular service request.

For the *Create User Account* Catalog Item, access was configured for new employees, managers, the Human Resources (HR) team, and the Identity and Access Management (IAM) team. For the *Enterprise VPN Access Request*, access was restricted to managers.

The User Criteria were associated with the corresponding Catalog Items through the access configuration of each item. The selected criteria were added to the **Available For** configuration, allowing ServiceNow to determine which users can access the service.

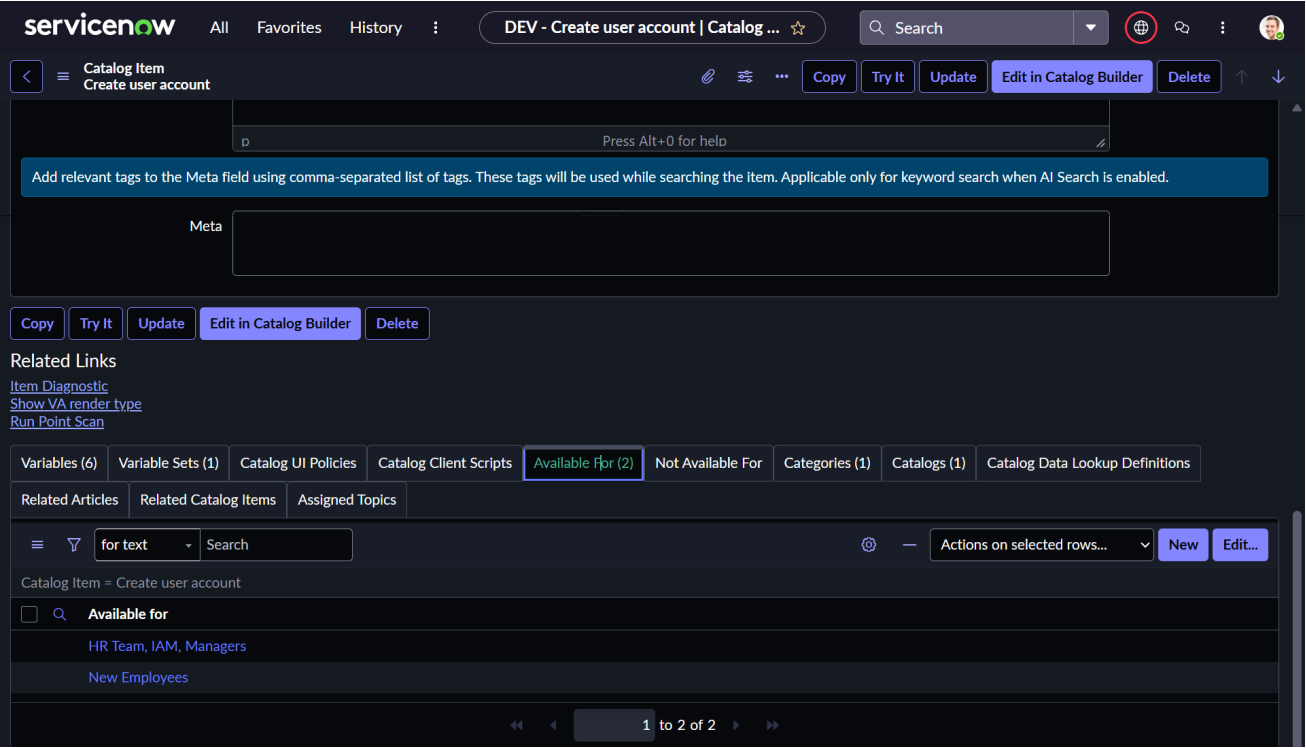


Figure 4.5: Associating User Criteria with a Catalog Item (Create User Account)

Figure 4.6 presents the User Criteria records configured for the different access requirements.

User Criteria Records									
Updated Search									
All									
☐	Name	Companies	Departments	Groups	Locations	Users	Roles	Match All	Updated
☐	vpn access Criteria			Managers		System Administrator		false	2026-08-18 10:28:05
☐	HR Team, IAM, Managers			HR Team, IAM, Managers		System Administrator		false	2026-08-15 13:14:31
☐	New Employees							false	2026-08-15 13:11:15

Figure 4.6: Configured User Criteria

A custom scripted User Criteria was also configured for new employees. This criterion dynamically evaluates the user's information to determine whether the user meets the defined conditions.

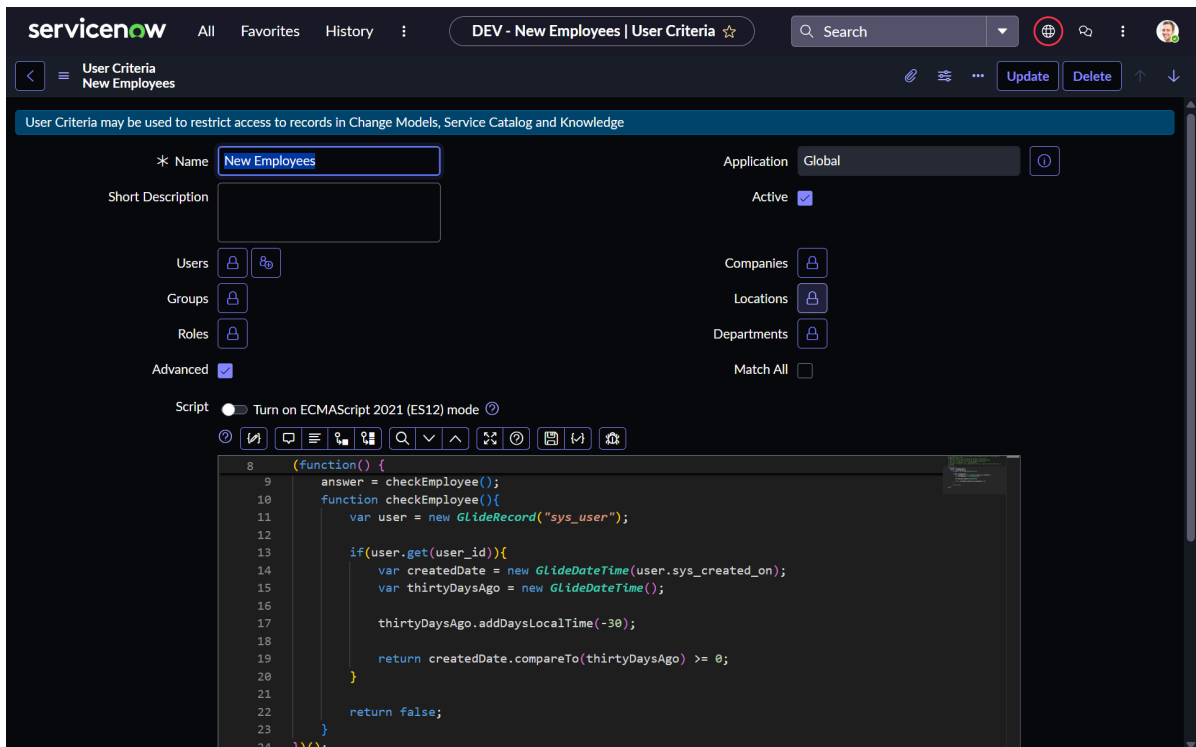


Figure 4.7: Custom User Criteria for New Employees

This configuration ensures that each Catalog Item is accessible only to the users who meet its defined access criteria.

4.4 Workflow and Process Automation

4.4.1 Flow Designer Configuration

The fulfillment processes of the Catalog Items were implemented using **Flow Designer**, ServiceNow's process automation tool. Flow Designer was used to automate the different stages of request processing, from the initial submission to approval, fulfillment, and closure.

Each Catalog Item was associated with a dedicated flow designed according to its specific requirements. The flows use triggers, actions, and conditional logic to determine the appropriate processing path for each request.

The general structure of the implemented flows consists of the following stages :

- **Trigger:** starts the flow when a request is submitted.
- **Get Catalog Variables:** retrieves the variables and values submitted through the

Catalog Item form, making the request information available to the subsequent flow actions.

- **Approval:** sends the request to the appropriate approver when approval is required.
- **Conditions:** evaluates request information such as account type, urgency, license availability, or equipment availability.
- **Catalog Tasks:** creates fulfillment tasks and assigns them to the appropriate teams.
- **Notifications:** informs users or teams when an action is required or when the request reaches an important stage.
- **Closure:** completes the request once the required fulfillment activities have been successfully performed.

The flows were configured progressively according to the requirements identified during the analysis and design phases. Although the overall structure is similar between the Catalog Items, the conditions, approvals, and fulfillment activities differ according to the requested service.

Figure 4.8 presents an example of a flow configured using Flow Designer.

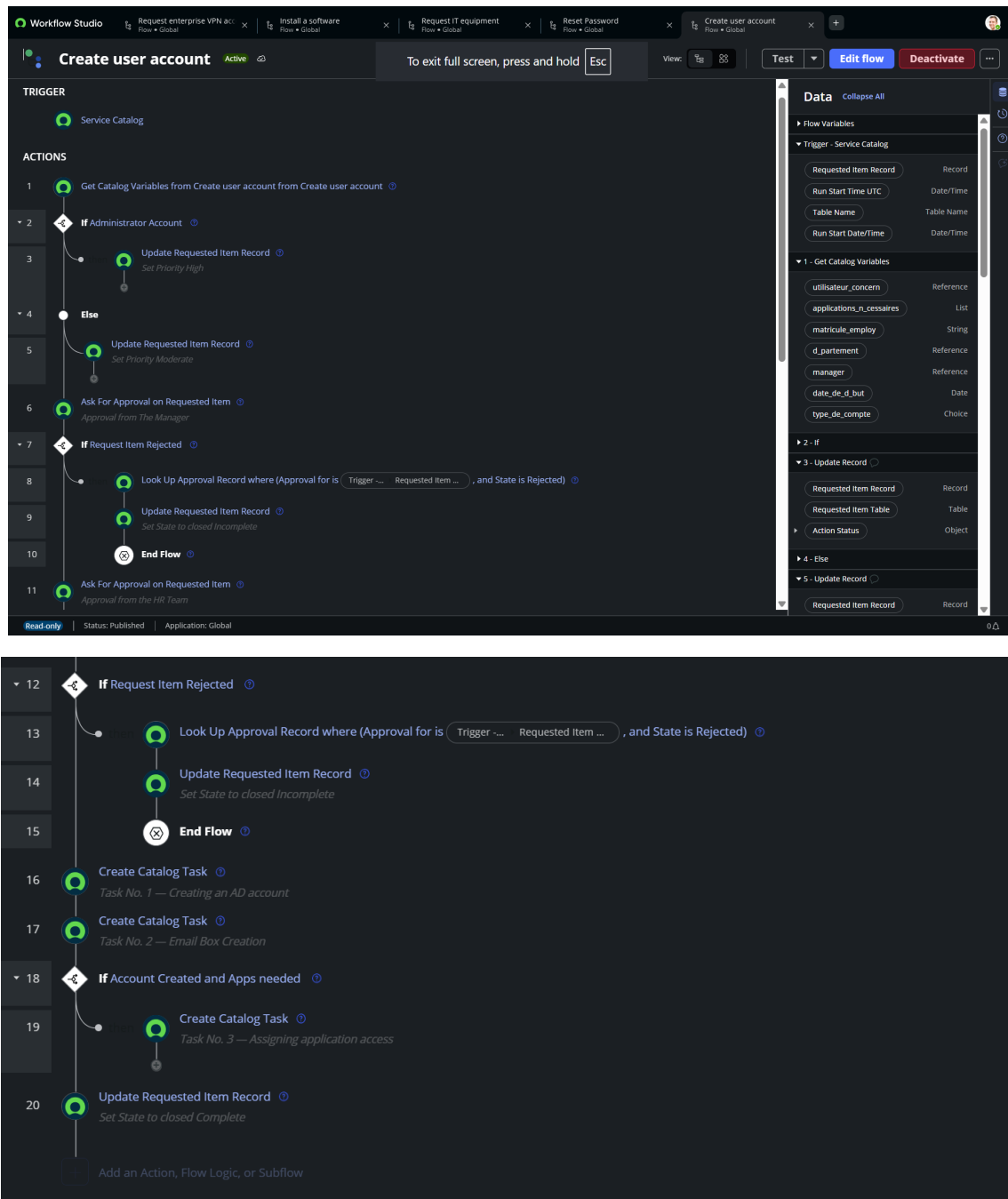


Figure 4.8: Flow Designer Configuration

4.4.2 Conditions and Branching

Conditional logic was used to adapt the fulfillment process according to the information provided in the request. The conditions configured in Flow Designer determine which path is followed and which actions are executed.

For example, the processing path may vary according to the account type, request urgency, software installation result, license availability, or equipment stock availability.

These conditions allow the workflow to automatically adapt to the characteristics of each request without requiring manual intervention for every decision.

Conditional branches were also used to handle specific verification and processing steps. In the *Software Installation* process, the user is asked to validate whether the requested software has been successfully installed and configured. If the validation is successful, the Requested Item is updated to *Closed Complete*. If the installation is reported as failed, the workflow follows an alternative path and updates the Requested Item to *Closed Incomplete*.

Conditions were also used to determine the priority of certain requests. The resulting priority can then be used by the applicable SLA definition to determine the expected fulfillment target.

Figure 4.9 illustrates the conditional branching used for user validation in the Software Installation workflow.

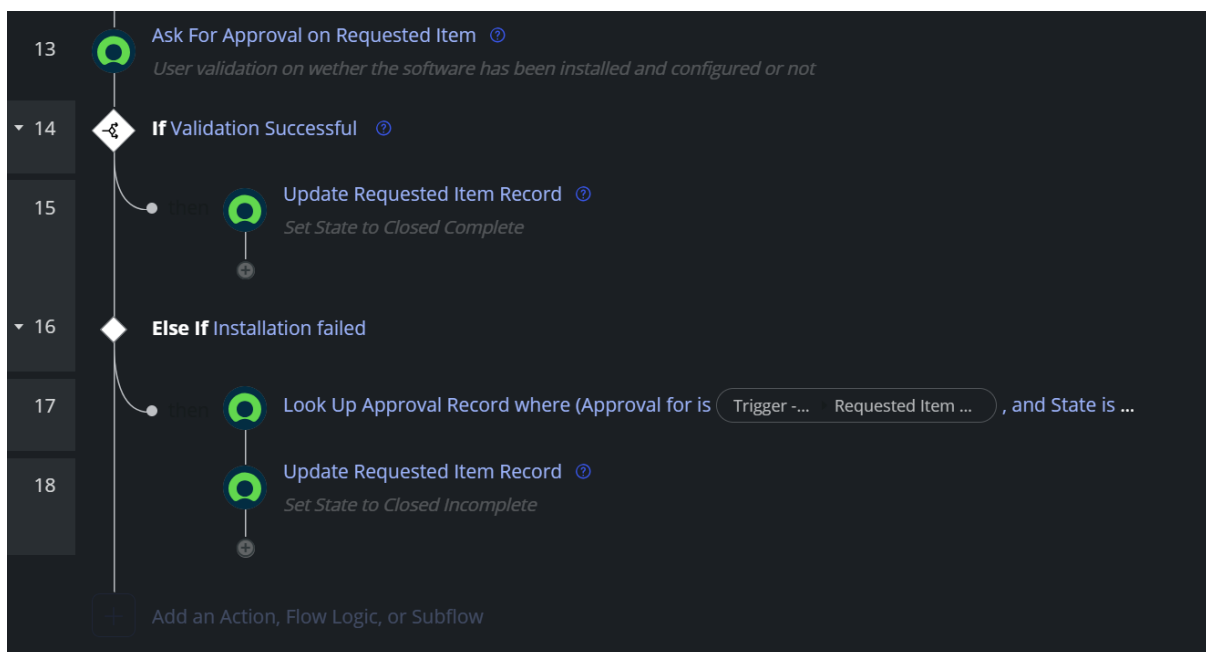


Figure 4.9: Conditional Branching for Software Installation Validation

4.4.3 Approvals and Rejection Handling

Approval steps were implemented in Flow Designer to ensure that requests requiring authorization are reviewed before the fulfillment activities begin. The approval process varies according to the type of service requested and the responsibilities associated with it.

For the *Create User Account* Catalog Item, the request is submitted for approval by the user's manager and then by the Human Resources (HR) team. For the *Enterprise VPN Access Request*, the request is first sent to the manager and then to the Security team. Other Catalog Items use different approval configurations according to their requirements.

The approval actions were configured to wait for the required decision before continuing with the subsequent fulfillment steps. When an approval is rejected, the workflow follows a dedicated rejection path and updates the Requested Item accordingly, preventing the request from continuing through the normal fulfillment process.

This approach ensures that requests requiring authorization cannot proceed to fulfillment without the required approvals, while rejected requests are properly identified and closed.

Figure 4.10 presents an example of the approval process configured in Flow Designer.

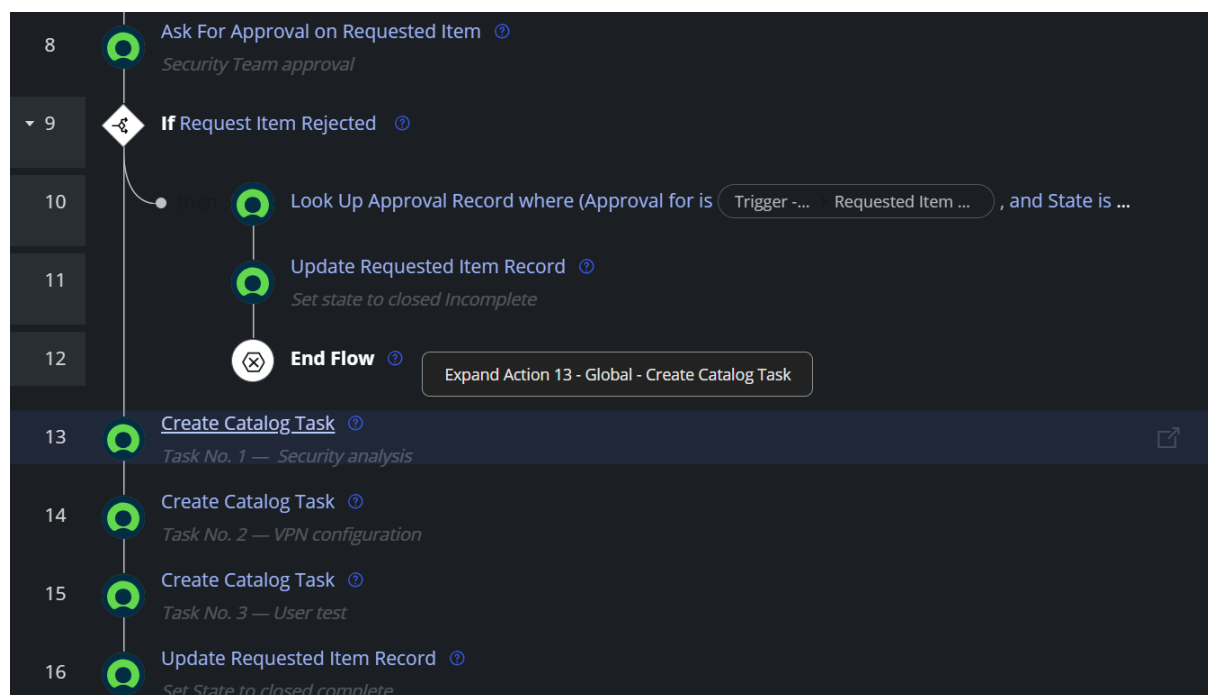


Figure 4.10: Approval Workflow For VPN Access Request

4.4.4 Catalog Tasks and Assignment

After the required approval steps and rejection handling, the workflows create the necessary **Catalog Tasks (SCTASKs)** to perform the fulfillment activities. Each task represents a specific operational activity required to complete the corresponding service request.

For each Catalog Task, an appropriate **Assignment Group** was configured according to the team responsible for performing the activity. The relevant Catalog Item variables were also made available to the fulfillment team, allowing them to access the information submitted by the requester directly from the task and use it during processing.

The Catalog Tasks and their corresponding Assignment Groups are presented in Table 4.2.

Catalog Item	Catalog Task	Assignment Group
Create User Account	AD Account Creation	IAM Group
Create User Account	Mailbox Creation	Messaging Team
Create User Account	Application Access	Application Support
Enterprise VPN Access Request	Security Analysis	Security Team
Enterprise VPN Access Request	VPN Configuration	Network Team
Enterprise VPN Access Request	User Testing	Service Desk
Software Installation	License Availability Check	Software Asset Management
Software Installation	Software Installation	Desktop Support
IT Equipment Request	Stock Check	IT Asset Management
IT Equipment Request	Equipment Preparation	Desktop Support
IT Equipment Request	Delivery	Logistics
Reset Password	Password Reset	Service Desk

Table 4.2: Catalog Tasks and Assignment Groups

The creation and assignment of these tasks were integrated directly into the corresponding Flow Designer workflows. This ensures that, once the required conditions and approvals are satisfied, the appropriate fulfillment team receives the task without requir-

ing manual routing.

Figure 4.11 presents an example of a Catalog Task configured in Flow Designer, including its assignment and the request information made available to the fulfillment team.

The screenshot displays the 'Create Catalog Task' configuration window in Flow Designer. The title bar indicates '16 Create Catalog Task' and 'Task No. 1 — Creating an AD account'. The 'Action Properties' section shows the 'Action' set to 'Create Catalog Task'. The 'Action Inputs' section includes several fields: 'Table Name' is 'Catalog Task [sc_task]'; 'Requested Item [Requested Item]' is 'Trigger ... Requested Item ...'; 'Short Description' is 'Tâche N° 1 — Création compte AD'; 'Assignment group' is 'IAM'; 'Fields' has a '+ Add field value' button; 'Wait' is checked; 'Template Catalog Item [Catalog Item]' is 'Create user account'; and 'Catalog Variables' shows two lists: 'Available' (empty) and 'Selected' (containing 'utilisateur_concern', 'matricule_employ', 'd_partement', 'manager', 'date_de_d_but', 'type_de_compte', and 'applications_n_cessa').

Figure 4.11: Catalog Task Configuration and Assignment

4.4.5 Notifications

Notifications were configured to keep the requester and the relevant users informed about important events during the fulfillment process. Three main notifications were implemented: one dedicated to the *Software Installation* process and two global notifications handling the closure of Catalog Items.

The first notification, *Software Installation User Validation*, is triggered when the software installation has been completed and the user's validation is required. The notification is associated with the approval record generated for the validation step and is sent to the corresponding approver. The email provides information about the installed software and the target machine, and includes actions allowing the user to validate the installation or report a problem.

Figure 4.12 presents the configuration of the notification used to request user validation.

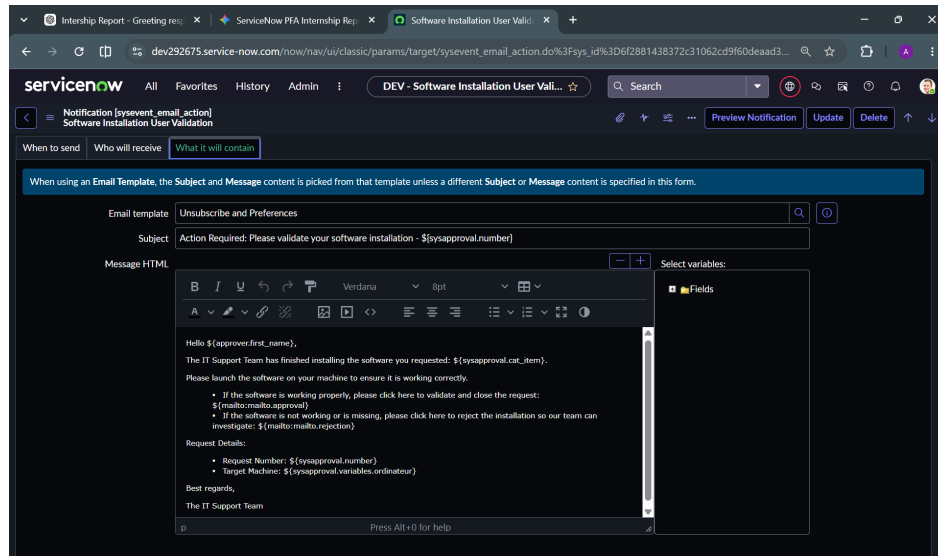


Figure 4.12: Software Installation User Validation Notification

Figure 4.13 presents a preview of the email sent to the user for software installation validation.

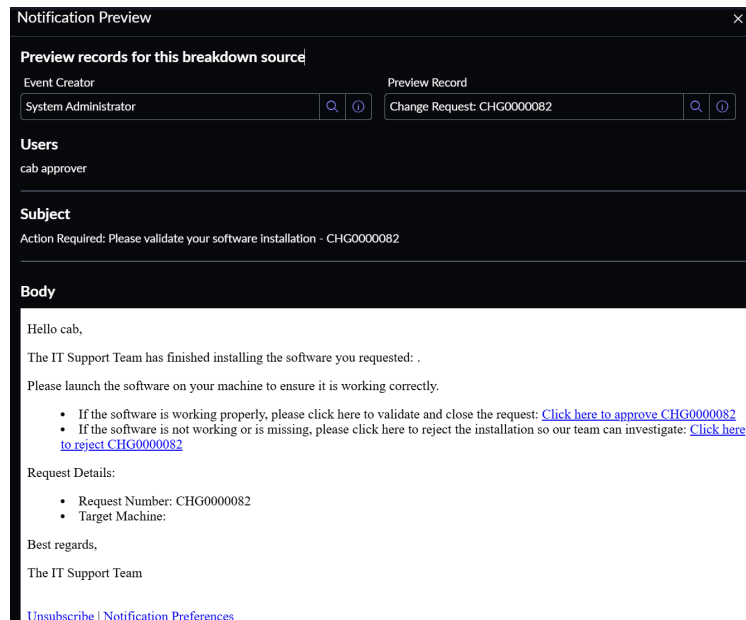


Figure 4.13: Software Installation User Validation Notification Preview

Two additional notifications were configured for the closure of Catalog Items. The first notification is triggered when a Requested Item reaches the *Closed Complete* state. It informs the requester that the service request has been successfully processed and closed.

The email includes information such as the request number, requested service, opening date, and closure notes.

Figure 4.14 shows the configuration of the notification for successfully completed requests.

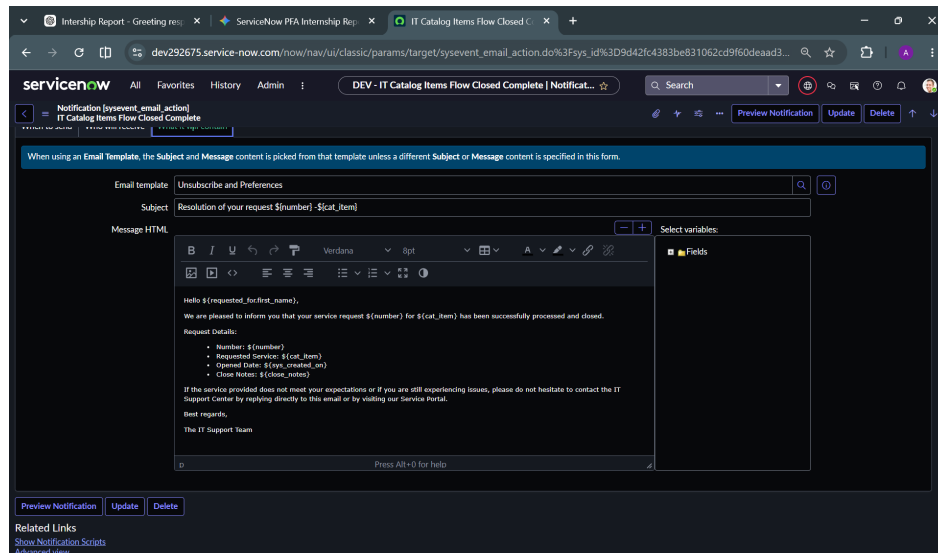


Figure 4.14: Closed Complete Notification Configuration

Figure 4.15 presents the email preview sent when a request is successfully completed.

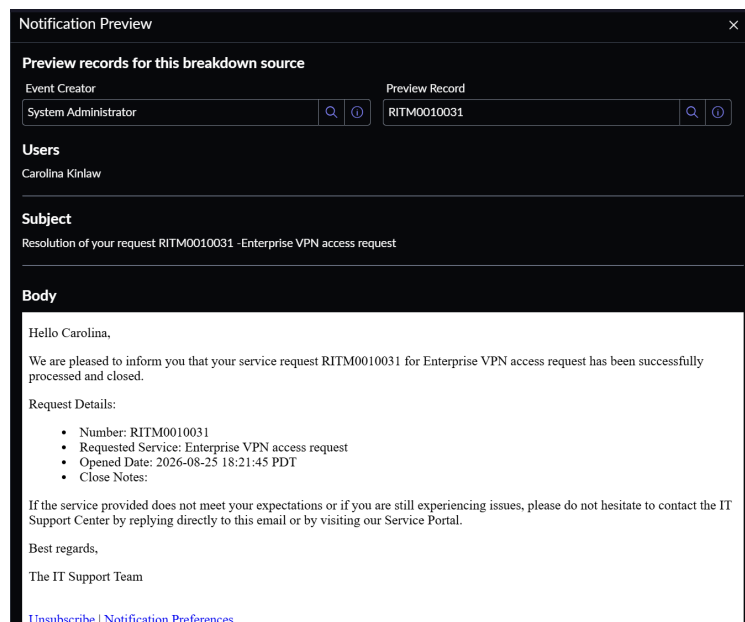


Figure 4.15: Closed Complete Notification Preview

The second global notification is triggered when a Requested Item reaches the *Closed*

Incomplete state. It informs the requester that the request could not be successfully completed and provides the available request details and rejection or closure information.

Figure 4.16 shows the configuration of the notification for requests closed as incomplete.

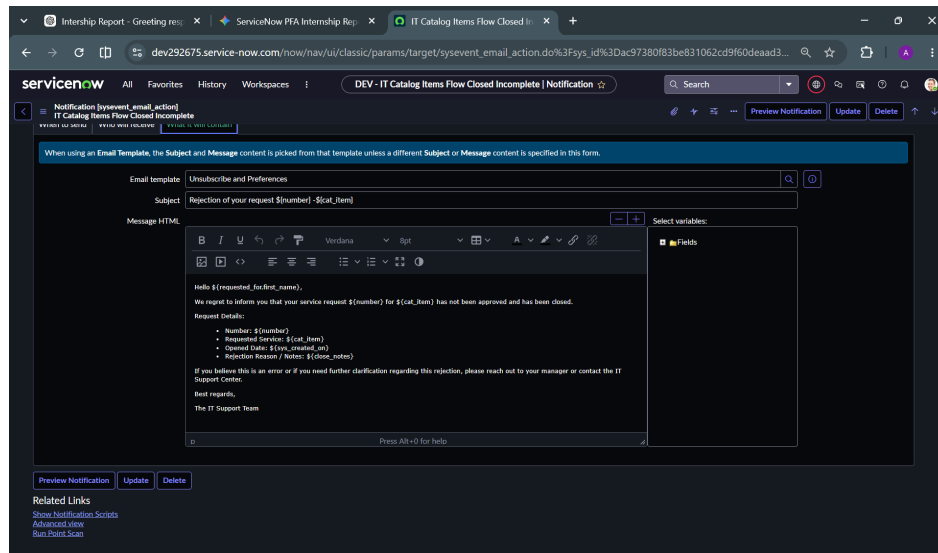


Figure 4.16: Closed Incomplete Notification Configuration

Figure 4.17 presents the email preview sent when a request is closed as incomplete.

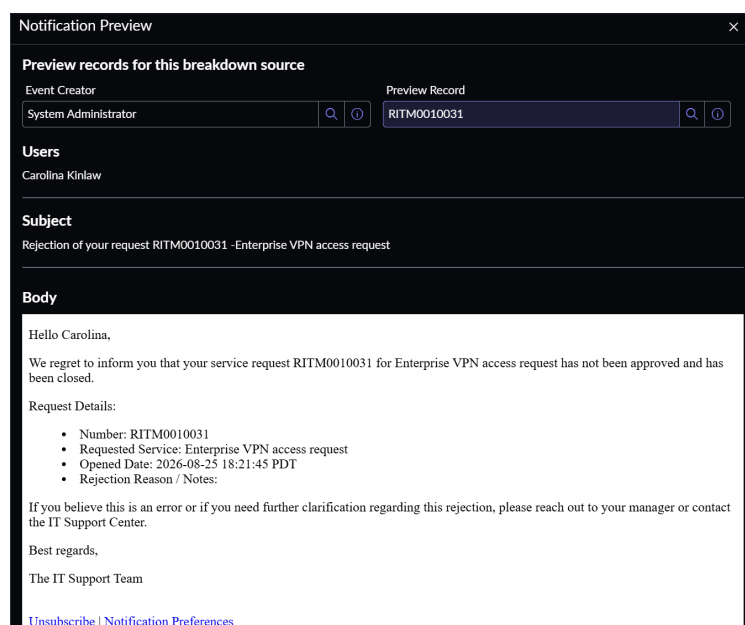


Figure 4.17: Closed Incomplete Notification Preview

For requests closed as incomplete, the workflow performs a lookup of the corresponding approval record before updating the Requested Item. This allows relevant information

from the approval record, such as the approver, to be retrieved and used during the subsequent processing of the request.

Figure 4.18 illustrates the lookup operation used in the workflow before the request is closed as incomplete.

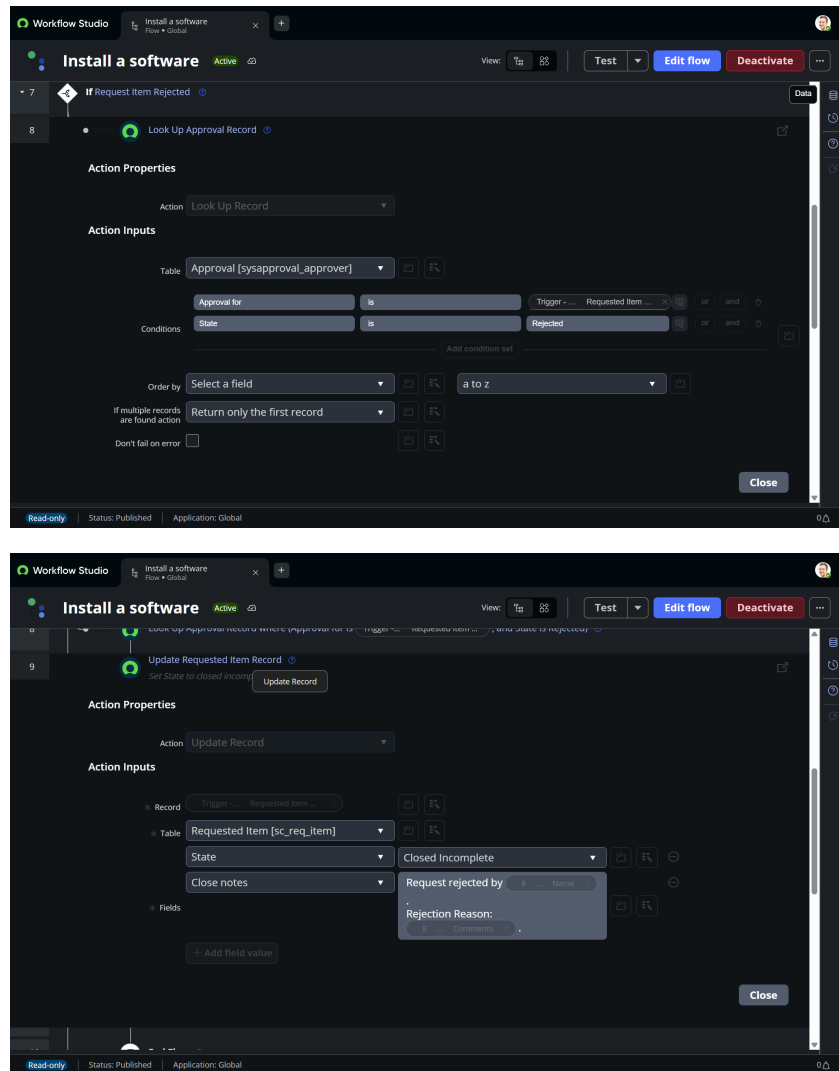


Figure 4.18: Approval Record Lookup Before Closing a Request as Incomplete

The notification configuration therefore provides communication at the main stages requiring user interaction or informing the requester of the final outcome of the service request.

4.5 Catalog Item Implementation

4.5.1 Create User Account

4.5.2 Enterprise VPN Access Request

4.5.3 Software Installation

4.5.4 IT Equipment Request

4.5.5 Reset Password